

# Teacher Guide — middle / module-10-herbs-aushadhi

IKS Curriculum

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## Overview

### Module 10 — Auṣadhi (Herbs)

**Age band:** Middle · **Grades:** 6–8 · **Duration:** 10 sessions × 45 min (≈ 2 weeks)

### Learning objectives

By the end of this module, students will be able to:

1. **Identify** six common South Asian herbs — *tulasī*, *nimba* (neem), *haridrā* (turmeric), *adraka* (ginger), *yavānī* (ajwain), *pudīnā* (mint) — by leaf shape, smell, and where appropriate the crushed-leaf scent.
2. **Match** each herb to its Sanskrit name, its botanical name, and the principal phytochemical compound modern chemistry has identified in it (eugenol, azadirachtin, curcumin, gingerol, thymol, menthol).
3. **Distinguish** culinary use (everyday, safe in food amounts) from therapeutic dosing (a clinical question for trained practitioners).
4. **Explain** how the *doṣa* framework from Module 4 sorts herbs into heating and cooling — *uṣṇa* and *śīta* — and predict which herb would traditionally be used for which kind of imbalance.
5. **Critically evaluate** a common claim about an Ayurvedic herb against evidence — articulating what is well-supported, what is preliminary, and what is folk-belief.

### NEP 2020 alignment

- **§4.6** — Integration of Indian Knowledge Systems with mainstream biology and chemistry.
- **§4.23** — Experiential learning through plant identification, smell-tests, and a kitchen-garden activity.
- **§11.8** — Multidisciplinary linkage (botany + chemistry + ethnobotany + Sanskrit terminology).
- **§4.27** — Local context: students learn herbs that grow in their kitchen, garden, or balcony.

## Prerequisites

- **Helpful (not strictly required):** Module 4 — Doṣas. The heating/cooling discussion on Day 8 lands faster if students already know *vāta*, *pitta*, *kapha*.
- Class 6 / 7 biology: parts of a plant; the basic idea that plants make chemicals.
- No Sanskrit knowledge assumed.

## Key vocabulary

See `_shared/glossary.md`. New for this module:

- *auṣadhi* — herb / medicinal plant
- *tulasī*, *nimba*, *haridrā*, *adraka*, *yavānī*, *pudīnā* — the six herbs we'll meet
- *tikta*, *kaṭu* — bitter, pungent (two of the six tastes; from Module 4)
- *uṣṇa*, *śīta* — heating, cooling (used to classify herbs)

## Safety framing — read before teaching

This module teaches students to **recognise** herbs and to understand how they are used. It does **not** teach prescription, dosing, or therapeutic use. The repeated message throughout is:

“These plants are safe in the small amounts people put in food. They are not safe in concentrated medicinal doses without a doctor’s supervision.”

Specific cautions covered in the lessons:

- **Neem** is intensely bitter and has documented liver toxicity (hepatotoxicity) at high or concentrated doses. Smell only. Do **not** chew the leaves in class. Do **not** ingest neem oil ever.
- **Essential oils** are not for ingestion at any age covered in this curriculum.
- **Pregnancy, children, ongoing medication** — dosing differs significantly and is a clinical question. Reinforce that this is not what we are teaching.
- **Allergy check** — before any smell or taste activity, ask the class about plant allergies and asthma triggers. Students with asthma may need to skip crush-and-sniff steps.

## Materials checklist (whole module)

- **Live plant samples (ideal):** tulsi plant in a pot, fresh ginger root, fresh turmeric root (or powder if root unavailable), ajwain seeds (and a small ajwain leaf cutting if accessible), fresh mint sprig, dried neem leaves (do not let students chew).
- **Backup:** clear glass jars of each as dried sample with a printed label.
- A4 paper for the herb identification card-deck (Day 9 project)
- Crayons / colour pencils for leaf-shape sketching
- Small sticky labels
- Index cards (~60) for the matching activity
- Optional: a kitchen-garden seed packet (tulsi, mint, coriander) for the homework / take-home

## Day-by-day overview

| Day | Title                                         | Focus                            | Assessment           |
|-----|-----------------------------------------------|----------------------------------|----------------------|
| 1   | What is an <i>auṣadhi</i> ?                   | Setting the stakes; meet the six | —                    |
| 2   | <i>Tulasī</i> — the household herb            | Identify, smell, eugenol         | Exit ticket          |
| 3   | <i>Nimba</i> — the bitter teacher             | Identify, taste-warning, safety  | Formative quiz       |
| 4   | <i>Haridrā</i> — golden root, modern molecule | Root vs powder; curcumin         | Reflection slip      |
| 5   | <i>Adraka</i> — fresh vs dried                | Gingerol vs shogaol; ginger tea  | Mid-module formative |
| 6   | <i>Yavānī</i> — seeds for the stomach         | Thymol; carom-water demo         | Exit ticket          |
| 7   | <i>Pudīnā</i> — cooling green                 | Menthol; varieties; kitchen uses | —                    |
| 8   | Doṣa lens — heating vs cooling herbs          | Link to Module 4                 | Concept map          |
| 9   | Project Session — build the herb deck         | Card-deck construction           | Project draft        |
| 10  | Showcase + summative                          | Card-deck presentations + quiz   | Summative + project  |

## Citation sources used

- ***Auṣadhi-sūkta*** — Ṛgveda 10.97 — the early Vedic hymn addressing herbs as healing beings. Referenced by sūkta number in Day 1; quoted as a short paraphrase in Day 8.
- ***Caraka Saṁhitā*, *Sūtrasthāna*** — the foundational Ayurvedic text where the *doṣa*-herb framework is laid out. Referenced at the chapter level only.
- Modern phytochemistry — referred to generically (“modern reviews indicate...”). The compound names (eugenol, curcumin, gingerol, thymol, menthol, azadirachtin) are basic chemistry-textbook level and not made up.

See `sources.md` for full detail.

## Project brief (one-line teaser)

Each pair builds a **6-card herb identification deck** — one card per herb — with: drawing of the leaf, Sanskrit + botanical name, three identifying features, the active compound, one traditional use, one safety note. Decks are presented on Day 10 and become a permanent classroom resource.

Full brief in assessments/project-brief.md.

## 10-Day Lesson Plan

### Module 10 (Middle) — 10-Day Lesson Plan

#### At a glance

| Day | Session title                                 | Lesson file                       | Activity / assessment                        |
|-----|-----------------------------------------------|-----------------------------------|----------------------------------------------|
| 1   | What is an <i>Auśadhi</i> ?                   | days/day-01-what-is-aushadhi.md   | Six-herb walkaround                          |
| 2   | <i>Tulasī</i> — The Household Herb            | days/day-02-tulasi.md             | Smell-and-sketch; exit ticket                |
| 3   | <i>Nimba</i> — The Bitter Teacher             | days/day-03-nimba.md              | quizzes/formative.md Part A                  |
| 4   | <i>Haridrā</i> — Golden Root, Modern Molecule | days/day-04-haridra.md            | Golden-milk preparation demo                 |
| 5   | <i>Adraka</i> — Fresh vs Dried                | days/day-05-adraka.md             | Ginger-tea demo; quizzes/formative.md Part B |
| 6   | <i>Yavānī</i> — Seeds for the Stomach         | days/day-06-yavani.md             | Ajwain-water demo                            |
| 7   | <i>Pudīnā</i> — The Cooling Green             | days/day-07-pudina.md             | Mint variety taste-id                        |
| 8   | Doṣa Lens — Heating and Cooling               | days/day-08-dosha-lens.md         | activities/activity-02-sort-by-virya.md      |
| 9   | Project Session — Build the Herb Deck         | days/day-09-project-session.md    | activities/activity-03-deck-build.md         |
| 10  | Showcase + Summative                          | days/day-10-showcase-summative.md | quizzes/summative.md + project               |

#### Pacing notes

- Days 2–7 are one herb per day. If you only have 6 sessions total, drop Days 7 and 9 (combine mint into the Day 8 doṣa lens; rush the project into Day 10 itself).
- Day 4 and Day 5 both have a short food-preparation demo. **Check school policy on cooking-in-class before the module starts.** If not allowed, both demos can be replaced with a smell-only station — and the “golden milk” / “ginger tea” become take-home recipes.

- Activity 02 (sort by *vīrya*) needs the *doṣa* vocabulary from Module 4. If your students haven't taken Module 4, allocate 10 min on Day 8 to introduce *uṣṇa* / *śīta* / *vāta* / *pitta* / *kapha* before the sort.

## Suggested running theme

Open each class with the same prompt on the board: **“Which herb did you smell or eat at home since our last class?”** Two students share for 30 seconds each. By Day 10 most students will have actively noticed the herbs already living in their kitchens.

## Pacing for the safety reminder

The **“these are not prescriptions”** line gets repeated **at the start of every herb-day lesson** (Days 2–7). Yes, every one. It is the most important takeaway of the module. Students should be able to recite it back by Day 5.

## Formative vs. summative weight

- Formative quizzes (Day 3 Part A; Day 5 Part B) — **not graded**. Use them to decide who needs the pre-class refresher on Day 6.
- Summative quiz (Day 10) is closed-book, 20 min, 25 marks. Combined with the project rubric (15 marks), the module total is 40 marks.

## What students leave the module with

- The ability to identify six common herbs by sight + smell.
- One project artefact (the 6-card herb deck) that they can take home or leave for the classroom library.
- The habit of asking “*what’s in this?*” and “*is it safe in this amount?*” before believing a wellness claim.
- The understanding that traditional knowledge and modern phytochemistry are **complementary**, not competing.

## Teacher Notes

### Teacher Notes — Module 10 Auṣadhi (Middle)

#### Posture for this module

Plants make chemicals. Many of those chemicals have effects on the human body. People in many cultures have observed those effects over long timescales and built systems of use. Some of those uses are well-supported by modern pharmacology; some are still being tested; some are folk-belief.

Teach all three honestly. The framing for this module is **observational botany meets modern chemistry meets cultural memory** — and the most important habit you can leave students with is the habit of asking “*is this safe in this amount?*” before reaching for any herb-based remedy.

If a student says “*my grandmother says turmeric milk cures everything*” — that’s a starting point, not a wrong answer. Acknowledge the lineage, ask what specifically is being claimed, then ask what evidence we’d want to evaluate the claim.

## The one safety message (repeat daily)

**Food amounts: safe.** A pinch of turmeric in your daal, a leaf of mint in your raita, a slice of ginger in tea — these are how millions of people have safely used these herbs for centuries.

**Therapeutic doses: a clinical question.** Powders, capsules, oils, extracts, or “take 3 tsp twice a day” — that is *medicine*, and medicine needs a trained practitioner. We do **not** teach prescription in this module.

Read this aloud on Day 1 and at least once on every herb-day (2 through 7). If you only have time for one thing on the safety side, choose this.

## Specific safety points

1. **Neem.** Do **not** let students chew the leaves. The bitterness is real and unpleasant, and concentrated neem (oil, extracts, large quantities of leaves) is **hepatotoxic** in well-documented case reports. Smell only. The seed-derived neem oil is *agricultural* — for pest control. It is **not** for ingestion.
2. **Essential oils.** Period. We do not put essential oils into bodies in this module. Not on skin, not in mouths, not in noses. They are highly concentrated and not appropriate for any school activity.
3. **Pregnancy.** If a student mentions a family member who is pregnant and considering an herb, redirect: “That’s a question for the doctor or a qualified practitioner — not for the class.” Some herbs studied here (notably neem oil, and some preparations of ginger at high doses) have known pregnancy contraindications. Don’t speculate.
4. **Allergy check.** Before Day 2, send a parent note asking about plant allergies. Even tulsi (*Lamiaceae*) can trigger reactions in students with severe pollen or aromatic-plant allergies.
5. **Asthma triggers.** Crushed-leaf smell tests may bother asthmatic students. Offer them a viewing-only role; don’t single them out.
6. **Hot drinks.** Ginger tea and golden milk should be lukewarm when students sip. Pre-cooled.

## Common misconceptions to watch for

1. **“Natural = safe.”** This is the single most common error. Poison ivy is natural. Cobra venom is natural. The relevant question is “*safe in what amount, for whom, in what preparation?*” — not whether a substance came from a plant.
2. **“If a little is good, more is better.”** No. Many plant compounds have **non-monotonic dose responses** — small amounts are beneficial, larger amounts toxic. Curcumin at culinary doses is well-tolerated by most people; at multi-gram daily doses it can cause stomach upset and, in rare cases, liver issues.
3. **“Modern science has proved Ayurveda works.”** Modern science has produced evidence for some specific compounds (curcumin’s anti-inflammatory mechanism is

well-studied) — but “Ayurveda” as a whole is a system of practice that includes diagnosis, individual constitution, lifestyle. You cannot validate or invalidate the whole system with one molecule.

4. **“Modern science has proved Ayurveda doesn’t work.”** Same misconception in reverse. Absence of a positive trial for one preparation does not invalidate centuries of empirical use.
5. **“Ayurvedic herbs are interchangeable with their named modern molecule.”** Whole-plant preparations contain hundreds of compounds. Isolating “the active ingredient” sometimes loses or gains effects compared to the whole.
6. **Confusing fresh and dried ginger.** *Adraka* (fresh) and *sūnṭhī* (dried) are the same plant, but Ayurveda treats them differently. Modern chemistry agrees: gingerol (fresh) and shogaol (dried) have different properties.

## Linking back to Module 4 (Doṣas)

Day 8 leans on the *vāta* / *pitta* / *kapha* framework. If your class hasn’t taken Module 4:

- Give 10 minutes at the start of Day 8 to introduce: three functional principles (motion, transformation, structure), each linked to two elements.
- Stick to the **heating/cooling (uṣṇa/śīta)** dimension for Day 8 — it’s the easiest entry.
- Skip the more complex *rasa-vīrya-vipāka-prabhāva* fourfold which belongs at Senior band.

## What “evidence-based” means at Middle band

At Middle band, distinguish three levels of claim:

1. **Folk use** — “my grandmother gives ginger tea for upset stomach.” Real human practice, not yet evidence-based in the trial sense.
2. **Mechanism evidence** — “gingerol activates TRP receptors and stimulates digestive enzyme release.” Lab-level evidence; useful for explaining mechanism but doesn’t directly establish clinical effect.
3. **Clinical evidence** — “in a randomised trial of 200 people, ginger reduced nausea more than placebo.” Trial-level evidence.

For most herbs in this module, you’ll have folk use ✓, some mechanism evidence ✓, and **limited or mixed clinical evidence**. Be honest about the gap. The honesty itself is the teaching.

## Sanskrit pronunciation

- *auśadhi* — “OW-shu-dhi” (the *au* as in “now”)
- *tulasī* — “TU-luh-see” (the final *ī* is long)
- *nimba* — “NIM-buh”
- *haridrā* — “huh-rid-RAH”
- *adraka* — “UD-ruh-kuh”
- *yavānī* — “yuh-VAH-nee”
- *pudīnā* — “pu-DEE-naa”

- *uṣṇa* — “OOSH-na” (heating)
- *śīta* — “SHEE-tuh” (cooling)

## Differentiation hints

- **Tier up (fast finishers):** Have them research a **seventh** herb not covered in the module — *aśvagandhā* (*Withania somnifera*), *brāhmī* (*Bacopa monnieri*), or *triphalā* (the three-fruit combination) — and add a 7th card to their deck. They must apply the same evidence-rigor: traditional use, modern compound, safety, evidence level.
- **Tier down (scaffolding):** Reduce to four herbs (tulsi, ginger, turmeric, mint) — the most commonly encountered in Indian kitchens. Add neem and ajwain as bonus cards if time permits.

## Sensitivity notes

- **Religious framing.** Tulsi is the only herb in this module with strong devotional associations in some households. Acknowledge respectfully: “many homes treat the tulsi plant as sacred — we are studying it here as a plant with specific chemical properties; both framings can coexist.”
- **Health claims at home.** Students may come in with strong claims from older relatives. Treat the relative respectfully; engage with the specific claim rather than the source.
- **Caste-and-Ayurveda discussion.** Don’t open this at Middle band. If a student raises it, acknowledge it briefly (“yes, like many ancient knowledge systems, Ayurveda was historically transmitted through specific social channels — this is one reason we study it now in a school open to all”) and move on.

## Project oversight (Days 9–10)

- On Day 9, every pair must have **all six cards started** by end of class. If a pair has only 2–3 cards drafted, intervene.
- The deck is small enough that it doesn’t need a “rough draft → final” iteration. One pass with care is enough.
- Show the rubric to students at the start of Day 9, not Day 10.

## If you only have 6 days, not 10

Compress: - Day 1 = Day 1 (intro) - Day 2 = Days 2 + 3 (tulsi + neem) - Day 3 = Days 4 + 5 (turmeric + ginger) - Day 4 = Days 6 + 7 (ajwain + mint) - Day 5 = Days 8 + 9 (doṣa lens + start deck) - Day 6 = Day 10 (showcase + quiz)

Drop the “smell-test challenge” if compressed; keep the safety message.

## After this module

- **Module 11 (Yoga therapy)** picks up the safety/individuality framing.
- **Module 13 (Ecology)** picks up the sustainability question: which of these herbs grow easily at home, which are stressed in the wild (neem is widely cultivated; some Ayurvedic herbs *not* in this module, like *aśvagandhā* and many forest herbs, are under sustainability pressure).



# Sources

## Sources — Module 10 Auṣadhi (Middle)

### Primary scriptural / classical sources

| Citation                                        | Used in      | Content                                                                                                                                                                                            |
|-------------------------------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ṛgveda 10.97</b> (the <i>Auṣadhi-sūkta</i> ) | Day 1, Day 8 | The hymn addressing the medicinal herbs as healing beings. Cited at the sūkta level only; a short paraphrase appears on Day 8.                                                                     |
| <b>Caraka Saṁhitā, Sūtrasthāna 1</b>            | Day 1, Day 8 | Foundational Ayurvedic text introducing the doṣa framework and <i>dravya-guṇa</i> (substance-quality) principles that justify the heating/cooling classification. Cited at the chapter level only. |

The RAG database was unavailable during build, so we do not include verbatim verse quotations beyond what is in the public-domain Ṛgveda. The *Auṣadhi-sūkta* paraphrase used on Day 8 reads:

“May the herbs that grew before the gods, brown and bright, born of past ages — may they protect the one who calls on them.”

— paraphrase of the opening of Ṛgveda 10.97 (*Auṣadhi-sūkta*)

This is a paraphrase, not a translation; teachers wishing to quote a published translation should consult Griffith (1896) or Jamison & Brereton (2014, Oxford UP) — both real published translations.

### Modern phytochemistry references

These compound names are basic chemistry-textbook level. We have **not** cited specific journal papers because we were unable to verify exact titles against the RAG. Treat the following as orienting reading for the teacher:

- **Curcumin** and turmeric — broadly reviewed in journals such as *Foods*, *Nutrients*, and the *Journal of Ethnopharmacology*. Review papers note its low bioavailability when consumed alone and improved absorption when paired with piperine (black pepper).
- **Gingerol** and **shogaol** — the active compounds in fresh and dried ginger respectively. Fresh ginger contains [6]-gingerol; drying converts much of it to [6]-shogaol, which is more pungent.

- **Eugenol** — the principal volatile aromatic in *Ocimum tenuiflorum* (tulsi) and also in clove (*Syzygium aromaticum*).
- **Thymol** — the principal volatile in ajwain seed; shared with thyme (*Thymus vulgaris*).
- **Menthol** — the principal volatile in *Mentha* species.
- **Azadirachtin** — the principal limonoid in neem; well known in agriculture as a natural insect repellent.

## Modern Indian regulatory sources (teacher reference only)

- **The Ayurvedic Pharmacopoeia of India (API)**, published by the Ministry of AYUSH, Government of India. Real source. Provides monographs on standard Ayurvedic herbs including all six in this module.
- **Indian Council of Medical Research (ICMR)** — periodic reviews on Ayurvedic herbs and integrative medicine.

## NCERT cross-references (for Day 8)

- NCERT Class 6 Science, Chapter 7: “Getting to Know Plants” — leaf shapes and venation.
- NCERT Class 7 Science, Chapter 1: “Nutrition in Plants” — photosynthesis as the source of plant secondary metabolites (gives a foothold for the curcumin / eugenol discussion).

## Sensitivity note on sources

The Ṛgveda is widely available as a religious text and as a historical primary source. We use it here as the latter — a window into how a society 3000+ years ago thought about plants and healing. We do not require students to relate to it devotionally and do not present it as devotional content in class.

## Day 1 — What is an *Auśadhi*?

**Module:** Auśadhi · **Band:** Middle · **Time:** 45 min

## Learning objective

By the end of this lesson, students can define *auśadhi*, name at least four of the six herbs the module will cover, and articulate the core safety rule: **food amounts are safe; therapeutic doses are a clinical question.**

## Materials

- Six labelled samples laid out at the front of the class: tulsi sprig, neem leaf (dried, no chewing), turmeric root or powder, ginger root, ajwain seeds in a small jar, mint sprig
- Whiteboard
- Six index cards, one per herb, with the Sanskrit + English + botanical name pre-printed

## Vocabulary introduced today

- *auṣadhi* (IAST: auṣadhi; lit. “herb, plant used as medicine”) — the general term, in Vedic and Ayurvedic Sanskrit, for any plant valued for healing.
- *tulasī*, *nimba*, *haridrā*, *adraka*, *yavānī*, *pudīnā* — the six herbs in this module.

## Lesson flow

### Warm-up (5 min)

Hold up the tulsi sprig. “How many of you have seen this plant at home?” Hands.

“How many of you have eaten what’s in this jar?” (Pull out ajwain.) Hands.

Establish the premise: most of these are already in their kitchens. We’re going to study what’s actually in them — and how Indian tradition organised that knowledge.

### Core (20 min)

1. **What is an *auṣadhi*?** Write the word on the board.
  - *Auṣadhi* means “plant used for healing.” It appears in the Ṛgveda, in a hymn called the *Auṣadhi-sūkta* (Ṛgveda 10.97), where the herbs are addressed directly — as though they were beings with names and personalities. We’ll come back to this hymn on Day 8.
  - In *Caraka Saṁhitā*, *Sūtrasthāna* 1 — one of the foundational texts of Āyurveda — herbs are sorted by their qualities (*guṇa*), their tastes (*rasa*), their heating/cooling potency (*vīrya*), and their effect after digestion (*vipāka*). Don’t worry about all four right now — we’ll meet *vīrya* on Day 8.
2. **The six herbs we’ll study.** Walk along the front bench, holding up each sample:

| Sanskrit       | English           | Botanical                  | Where you’ve seen it    |
|----------------|-------------------|----------------------------|-------------------------|
| <i>tulasī</i>  | tulsi, holy basil | <i>Ocimum tenuiflorum</i>  | garden pot, prayer area |
| <i>nimba</i>   | neem              | <i>Azadirachta indica</i>  | street tree, soap       |
| <i>haridrā</i> | turmeric, haldi   | <i>Curcuma longa</i>       | daal, milk              |
| <i>adraka</i>  | ginger, adrak     | <i>Zingiber officinale</i> | tea, sabzi              |
| <i>yavānī</i>  | ajwain, carom     | <i>Trachyspermum ammi</i>  | parathas, achaar        |
| <i>pudīnā</i>  | mint, pudina      | <i>Mentha</i> species      | chutney, raita          |

Ask the class: “Which two of these are you most familiar with? Which are you not?”  
Show of hands per herb.

3. **The big safety rule** (write on board in capitals — leave it there for the rest of the module):

## FOOD AMOUNTS: SAFE. THERAPEUTIC DOSES: ASK A DOCTOR.

Explain in one minute: every plant on this table contains chemicals that affect the human body. In small amounts — the pinch you put in food — these chemicals are mostly safe and often helpful. In concentrated form — powders, capsules, oils — they become medicine, and medicine needs supervision.

*“Anyone here ever heard a relative say ‘neem is good for you, eat the leaves’? Does anyone want to chew a neem leaf right now?” (They will. Don’t let them.) “Neem in concentrated form can damage the liver — and pure neem leaf is so bitter you’d spit it out anyway. So we’ll smell it. We won’t eat it.”*

### Activity (15 min) — Six-herb walkaround

Set up six “stations” — one per herb. Pairs of students rotate through, 1.5 minutes per station. At each:

- **Look.** What shape is the leaf or root?
- **Smell.** Crush a tiny piece (where safe — not neem) and inhale gently from a distance. Don’t put it under anyone else’s nose.
- **Record** one identifying feature on their workbook page.

At the neem station, leaves are dried and in a sealed bag. **Smell from outside the bag.** No touching the inside.

Last 3 min: pairs report one feature back to the group. Build a quick “first impression” word-cloud on the board.

### Wrap-up (5 min)

- Restate the safety rule chorally — the whole class.
- Exit ticket on a slip: write the **Sanskrit name and one identifying feature** for any two herbs you remember.
- Homework: tonight, find as many of the six herbs as you can in your kitchen or balcony. List them.

### Student-facing content

An **auṣadhi** is a plant used for healing.

We’re going to study six of them — every one is already in most Indian kitchens or gardens:

*tulasī, nimba, haridrā, adraka, yavānī, pudīnā.*

**The rule:** food amounts are safe. Therapeutic doses (capsules, powders, oils) are for trained practitioners, not for us.

The early Indian tradition addressed these plants directly in a hymn called the *Auṣadhi-sūkta* (R̥gveda 10.97). Later, the *Caraka Saṁhitā* (one of the foundational Ayurvedic texts) sorted them by their tastes, qualities, and heating/cooling effect. We’ll meet both these ideas over the next two weeks.

## Homework

- List which of the six herbs are in your kitchen tonight. Bonus: bring a leaf or a piece (with a parent’s permission) to class tomorrow for the tulsi day.

## Differentiation

- **One tier up:** Find a *seventh* herb in your kitchen we haven’t named (cilantro, fennel, fenugreek, curry leaf, etc.). What might its Sanskrit name be? (Hint: look up *dhānyaka*, *methi*, *kariveppilai*.)
- **One tier down:** Just remember three herbs by Friday — tulsi, turmeric, ginger. The other three can come over the next week.

## Teacher notes

- Have the safety rule visible on the board for **every class** in this module, not just today.
- A few students will laugh at “addressed the herbs as if they were beings.” Use this. It’s a glimpse of a world where you talked to plants. We don’t have to take the metaphysics literally to take the *attention* seriously.
- The class word-cloud from the walkaround is your fastest read on who already knows what. Use it to plan Days 2–7.

## Citation(s) used in this lesson

- *R̥gveda* 10.97 — the *Auṣadhi-sūkta*. Referenced by sūkta number. Real public-domain Vedic source.
- *Caraka Saṁhitā*, *Sūtrasthāna* 1. Referenced at chapter level. Real foundational Ayurvedic source.

## Day 2 — *Tulasī* (Holy Basil)

**Module:** Auṣadhi · **Band:** Middle · **Time:** 45 min

## Learning objective

By the end of this lesson, students can identify *tulasī* (*Ocimum tenuiflorum*) by leaf shape and crushed-leaf smell, name eugenol as its principal aromatic compound, and describe two traditional culinary or domestic uses without making therapeutic claims.

## Materials

- A live tulsi plant in a pot (ideal); a fresh sprig if a plant isn’t available
- A second Lamiaceae plant for comparison if you can — sweet basil (*Ocimum basilicum*) is the easy comparison
- A clove (the spice — also rich in eugenol) for the smell-bridge
- Whiteboard
- Student notebooks

## Vocabulary introduced today

- *tulasī* (IAST: *tulasī*; lit. “the incomparable one”) — *Ocimum tenuiflorum*, the herb commonly called holy basil.
- *eugenol* — a phenolic aromatic compound. The strong, slightly clove-like smell of tulsi is largely from eugenol. Also the main aromatic in actual clove (*Syzygium aromaticum*).

## Lesson flow

### Warm-up (5 min)

Pass the tulsi plant slowly along one row. Each student takes a single leaf between their fingers, rubs it gently, and sniffs.

“What does it smell like? Three words.” Take 5–6 responses. Likely answers: “sharp,” “minty,” “spicy,” “clove-y,” “medicine-ish.”

Now pass the clove. “Familiar?” The connection is the molecule. Don’t tell them yet — let them notice.

### Core (20 min)

1. **Identify it.** Tulsi has:
  - **Leaf shape:** ovate (oval-pointed), about 2–5 cm long, with **toothed margins** (small jagged edges, called “serrate” in botany).
  - **Stem:** square in cross-section — characteristic of the **Lamiaceae** family (mint family). All six plants in this family share this. Press a stem between thumb and forefinger and feel the corners.
  - **Smell when crushed:** the sharp, clove-like aroma.
  - **Common varieties in India:** *Rāma tulasī* (green), *Kṛṣṇa tulasī* (purple-leaved). Same species, different cultivars.
2. **What’s inside.** Modern phytochemistry has identified the main aromatic in tulsi as **eugenol** — a phenolic compound ( $C_{10}H_{12}O_2$ ). The same compound dominates clove (the spice). That’s why the two smell related.

Why does the plant make eugenol? Because it’s the plant’s chemical defence — eugenol discourages many insects from eating the leaves, and it has antibacterial effects in petri-dish studies. The plant doesn’t make it for us; we just happened to find it useful.

3. **Traditional uses.** Mention only:
  - As a *kitchen herb* — in chutneys, drinks (tulsi tea — leaves steeped in water), sometimes a few leaves added to dal.
  - As a *household plant* — many Indian homes keep a tulsi plant for its scent and its insect-repelling qualities.
  - As an *aromatic* — in some traditional Ayurvedic preparations for cough and cold support. (Here, drop the safety reminder again — these preparations are formulated by practitioners; we are not recommending self-medication.)

4. **What we won't claim.** Tulsi is sometimes marketed in capsules as an “adaptogen” — a label that means “helps your body handle stress.” There is *some* mechanistic evidence (anti-inflammatory effects from eugenol and other compounds), but clinical trial evidence is **mixed**. Saying “tulsi reduces stress” as a blanket claim isn't honest; saying “people have used tulsi as a calming household scent for a long time and there is preliminary evidence of physiological effects” is more accurate.

### Activity (15 min) — Smell-and-sketch

- Each student gets a piece of paper and crayons.
- Draw the tulsi leaf — accurate to shape, edge, and size. Label: leaf shape (ovate), edge (serrate), family (Lamiaceae, square stem).
- Below the drawing, write one sentence: “*What I smelled when I crushed the leaf was \_\_\_\_\_.*”
- Below that, the chemistry box: **Active compound: eugenol.**

Optional: pass around the sweet basil (or any other Lamiaceae) for comparison. “*Different leaf? Different smell? They are the same plant family but different species.*”

### Wrap-up (5 min)

- Exit ticket on a slip: “*Write the Sanskrit name and the principal aromatic compound of today's herb.*”
- Restate the safety rule chorally.
- Preview Day 3: “*Tomorrow we meet a much less friendly tasting herb — neem.*”

## Student-facing content

**Tulasī** (*Ocimum tenuiflorum*) — known in English as holy basil, in Hindi as tulsi.

**How to identify it:** - Ovate (oval-pointed) leaf, about 2–5 cm long, with small jagged edges. - Square stem (a feature of the Lamiaceae family). - Strong, sharp, clove-like smell when you rub a leaf between your fingers.

**What's inside:** The main aromatic compound is **eugenol** — also the principal aromatic in actual clove. The plant produces it as a chemical defence.

**Traditional uses:** Kitchen herb (tulsi tea, chutney), household plant, ingredient in some Ayurvedic cold-and-cough preparations.

**What we don't say:** “Tulsi cures X.” Folk use and preliminary evidence are not the same as clinical proof.

## Homework

- Find a tulsi plant — your own pot, a neighbour's, a temple's, a school garden. Sketch its leaf in your workbook from life (not from a picture). Write one identifying feature you observed in person.

## Differentiation

- **One tier up:** Sweet basil (*Ocimum basilicum*) has the same genus as tulsi but different uses. What chemical does sweet basil contain that tulsi doesn't dominate in (Hint: linalool)? Why does Italian cuisine use sweet basil but not tulsi?
- **One tier down:** Just remember the leaf shape and the word "eugenol." Smell-recognition will come with repetition.

## Teacher notes

- If a student tells the class that tulsi is worshipped in their home, acknowledge respectfully: "Yes — many homes treat the tulsi plant as sacred. We're studying it here as a plant with specific chemistry; both framings can coexist."
- The clove-tulsi connection is the most memorable hook. Lean into it.
- Some students will react strongly to the smell of crushed tulsi (it can be quite sharp). Don't put it under their nose — let them choose their distance.

## Citation(s) used in this lesson

- General phytochemistry: eugenol as principal volatile in *Ocimum tenuiflorum* — this is established botanical-chemistry textbook content, not a contested claim. No specific paper cited.

## Day 3 — *Nimba* (Neem)

**Module:** Auśadhi · **Band:** Middle · **Time:** 45 min

## Learning objective

By the end of this lesson, students can identify *nimba* (*Azadirachta indica*) by leaf shape, describe its dominant taste as *tikta* (bitter), name azadirachtin as its principal active compound, and explain why concentrated neem is **not** safe for ingestion at any age.

## Materials

- A few dried neem leaves in a sealed container (smelling only — no chewing)
- A picture or — if you can — a neem twig with the characteristic compound leaf
- A small bottle of neem oil **kept by the teacher** (do not pass to students)
- A piece of neem-containing soap (commercial — most Indian general stores stock these) for the "what's it actually in" demonstration
- Whiteboard

## Vocabulary introduced today

- *nimba* (IAST: nimba; lit. "the bitter one") — *Azadirachta indica*, the neem tree.
- *tikta* (IAST: tikta; lit. "bitter") — one of the six tastes in Ayurveda; neem is the canonical *tikta* plant.



- *azadirachtin* — a complex limonoid compound. The most-studied bioactive in neem; the molecule behind neem’s use as an agricultural insect repellent.

## Lesson flow

### Warm-up (5 min)

Daily check-in: who has seen a neem tree near their home, school, or daily route? Hands.

“Anyone here ever been told ‘eat a neem leaf, it’s good for you’? Did you?” (Most who tried will say it tasted terrible.)

Bridge: today we explore why the body’s *first response* to neem — “spit it out” — is actually telling us something useful.

### Core (20 min)

1. **Identify it.** Neem is a tree, not a small herb:
  - **Compound leaf:** instead of a single blade, each “leaf” is made up of 9–15 small leaflets attached to a central stem. (Compare to tulsi yesterday — single simple leaves.)
  - **Leaflet shape:** narrow, curved (sickle-shaped), with serrate (toothed) edges.
  - **Smell when crushed:** strong, slightly garlic-like, mostly bitter.
  - **Where it grows:** native to South Asia; widely planted along streets, in temple compounds, and as a shade tree.
2. **The taste.** Pass the sealed container around for smell only.

**Today we do not chew the leaf.**

Neem is the textbook example of *tikta* (bitter taste) in Ayurveda. The bitter taste comes from a class of compounds called **limonoids** — the most-studied of which is **azadirachtin**. Bitter taste is the body’s warning signal — many plant toxins are bitter, and the bitter receptors evolved to make us *cautious* about new plants.

3. **What azadirachtin does.** This is real:
  - **Insecticidal/insect repellent:** azadirachtin disrupts insect moulting hormones. Hence neem oil is widely sold as an agricultural insect repellent — and that use is well-supported.
  - **Antimicrobial effects** at lab/petri-dish level.
  - **Traditional uses:** twigs as natural toothbrushes (*dātun*), neem-leaf paste on skin in some folk traditions, neem-based soaps and shampoos. These external/dilute uses are different from internal use.
4. **The safety warning — direct.** Write on the board:

⚠️ **Neem is hepatotoxic at high doses.** Concentrated neem (especially neem oil ingested) has documented cases of acute liver injury, particularly in children. Neem oil is **not** for eating, drinking, or putting in anyone’s mouth.

This is a real, documented safety concern in case-report literature. Neem leaf in small amounts (a few leaves brewed as a bitter tea in some traditions) is generally tolerated by adults — but it is **not** what we are teaching here, and self-prescription is not appropriate.

#### 5. What's it in, that we DO use?

- Neem-based soap, shampoo, toothpaste — fine for external use as marketed.
- Neem oil for plants — used as agricultural pest control. Read the label.
- Neem twigs as toothbrushes — a long-standing dental tradition with modern dentistry's mixed view (some plaque-reducing benefit; abrasive enamel concern if used too often).

### Activity (15 min) — Formative Quiz Part A + Compound-leaf sketch

**First 5 min — Formative Quiz Part A** (from quizzes/formative.md). Hand out the 5-question slip. Pencils only. Collect at minute 5.

**Next 10 min — Compound-leaf sketch.** - Each student draws a neem compound leaf from observation (or from the photo if no live sample). Label the rachis (central stem), the leaflets, the serrate edges. - Below the drawing, label: - **Sanskrit name:** *nimba* - **Botanical name:** *Azadirachta indica* - **Taste class:** *tikta* (bitter) - **Active compound:** azadirachtin - **Safety:** *not for ingestion in concentrated form*

### Wrap-up (5 min)

- Restate safety rule + the neem-specific warning chorally.
- Hand-back of Day 2 exit tickets if marked.
- Preview Day 4: “Tomorrow — the herb that turns everything gold.”

### Student-facing content

**Nimba** (*Azadirachta indica*) — known in English as neem.

**How to identify it:** - A tree, not a small plant. - **Compound leaf** — what looks like one “leaf” is actually made of 9–15 small leaflets arranged along a central stem. - Leaflets are narrow, slightly curved, with jagged edges. - Smell when crushed: bitter, slightly garlic-like.

**What's inside: Azadirachtin** — a limonoid compound. The molecule behind neem's well-supported use as an agricultural insect repellent.

**Taste class in Ayurveda:** *tikta* — bitter. Bitter taste is the body's warning signal for many plant toxins.

**Safety:** - External uses (soap, oil for plants, twig as toothbrush) — fine when used as labelled. - **Concentrated internal use — especially neem oil — is unsafe.**

Documented cases of liver toxicity in children. - We do not chew, drink, or eat neem in this class.

## Homework

- Find a neem tree near your home or school. Sketch a leaf in your workbook from life. Note where the tree is and how many you saw on your walk.

## Differentiation

- **One tier up:** Read up on the *Margosa oil poisoning* literature (margosa is another name for neem oil). Find one case-report finding. Be ready to share next class.
- **One tier down:** Focus on identifying the compound-leaf shape. The chemistry (azadirachtin) can come on Day 6 once you’ve seen more herbs.

## Teacher notes

- The “don’t chew the leaf” rule is critical. A few brave/curious students will try if leaves are accessible — keep the container sealed and store it where students can’t reach during break.
- The hepatotoxicity warning needs to land as **specific knowledge**, not vague caution. Use the word “liver” explicitly. Some students have grandparents who give them neem leaves “for blood purification” — the lesson is to be respectful and curious about the traditional use *while* being clear about the dose concern.
- If a student asks “but if azadirachtin is dangerous, why have people used neem for centuries?” — the answer is: dilute external use and very small dietary use have a long history of being tolerated; *concentrated* use (oil, multi-leaf daily intake in adults, anything in children) is where the harm shows up.

## Citation(s) used in this lesson

- *Azadirachta indica* and azadirachtin as principal limonoid: standard ethnobotanical and pharmacognosy textbook content.
- Neem oil hepatotoxicity: documented in published case reports in pediatric toxicology literature. We do not cite a specific paper here — teachers who want to deepen this should search “margosa oil hepatotoxicity case report” in PubMed.

## Day 4 — *Haridrā* (Turmeric)

**Module:** Auśadhi · **Band:** Middle · **Time:** 45 min

### Learning objective

By the end of this lesson, students can identify *haridrā* (*Curcuma longa*) as a rhizome (not a root), name **curcumin** as its principal pigment and active compound, distinguish whole-spice culinary use from concentrated supplement use, and describe one well-supported and one over-claimed use.

### Materials

- Fresh turmeric rhizome (the orange-coloured fingerlike root-like piece) — ideally with a fresh piece cut to show the inside

- A small bowl of turmeric powder
- A piece of white paper for the staining demo
- A small piece of black pepper or a peppermill (for the curcumin-piperine pairing discussion)
- Optional, for the demo: warm milk, a small saucepan, turmeric powder, a pinch of pepper, jaggery — to make a single demonstration “golden milk” (subject to school food policy)
- Whiteboard

## Vocabulary introduced today

- *haridrā* (IAST: haridrā; lit. “the yellow one”) — *Curcuma longa*, the turmeric plant.
- **Rhizome** — an underground horizontal stem (not a root). Distinguishes turmeric and ginger from carrot or beet.
- **Curcumin** — the polyphenol responsible for turmeric’s yellow colour and most of its studied biological activity.

## Lesson flow

### Warm-up (5 min)

Hold up the fresh turmeric rhizome. Slice a tiny piece — bright orange interior. Pass on a small plate so students can see.

“*Hands up — who used turmeric in something you ate yesterday?*” (Almost everyone.) “*Hands up — who knows that the bright yellow colour is one specific molecule?*” (Almost nobody.)

That’s today’s hook: a single molecule is responsible for almost everything turmeric is famous for.

### Core (20 min)

1. **Identify it — rhizome, not root.** Important distinction:
  - A **root** absorbs water and nutrients (carrot, beetroot).
  - A **rhizome** is an underground stem that stores starch and produces new shoots (turmeric, ginger, ginger lily, bamboo).
  - Turmeric rhizomes are fingerlike, with skin that’s pale brown to grey on the outside and bright orange on the inside.
  - The plant above ground has long, broad green leaves — similar to ginger and to canna lily.
2. **Powder vs whole.** Both forms are used:
  - **Fresh rhizome:** sliced into curries, pickled, sometimes juiced.
  - **Powdered (dried, ground):** the everyday Indian kitchen form — added to dal, sabzi, curries.
  - **Why the colour?** Curcumin and related curcuminoids — diarylheptanoid polyphenols. The molecule absorbs blue light and reflects yellow-orange.

3. **Staining demo (1 min).** Dab a small amount of fresh turmeric on white paper. Permanent stain. “*This is why turmeric on your shirt is a problem. Curcumin binds to fibres.*”
4. **What modern chemistry knows.** Real, well-supported:
  - **Anti-inflammatory mechanism:** curcumin inhibits several inflammation-related signalling molecules in lab studies. Mechanism evidence is strong.
  - **Low bioavailability:** here’s the catch — curcumin is **poorly absorbed** by the human gut on its own. Most of what you eat passes through unabsorbed.
  - **Piperine (black pepper) increases absorption** by roughly 20×, by inhibiting a liver enzyme that breaks down curcumin. This is why traditional preparations often combine turmeric with pepper. The tradition figured out the absorption problem long before the molecule was named.
5. **What’s claimed vs what’s solid.**
  - **Well-supported:** Turmeric is a safe culinary spice. It contributes to the colour and a mild flavour. Curcumin has anti-inflammatory activity in lab studies.
  - **Mixed evidence:** Curcumin supplements for arthritis pain — *some* clinical trials show modest benefit; others show no effect beyond placebo.
  - **Overclaimed:** “Turmeric cures cancer.” This is a marketing claim, not a clinical finding. Curcumin is being studied for various cancer-related effects in the lab, but jumping from “lab activity” to “cure” is exactly the leap we are training students *not* to make.
6. **Traditional use in Ayurveda.** *Haridrā* is *kaṭu-tikta* (pungent-bitter) in taste, *uṣṇa* (heating) in *vīrya*. Used in formulations for skin conditions (external paste), wound healing (external paste), and digestive support (small culinary amounts). The bright pigment is also used at weddings (*haldi* ceremony) — non-medicinal cultural use.

### Activity (15 min) — Golden-milk demonstration (or recipe-card if no cooking allowed)

**If cooking is permitted:** - Teacher prepares a single demo cup: 200 ml warm milk + ¼ tsp turmeric powder + tiny pinch black pepper + jaggery to taste. Warm gently (do **not** boil). - Don’t drink in class. **The point is the recipe, not the consumption.** Or, prepare a cooled small tasting cup and offer to those without dairy/curcumin/dietary concerns.

**If cooking is not permitted (or as a complement):** - Each pair writes out a recipe card: ingredients, quantities, why each is there. - Specifically: *why pepper?* (bioavailability) *why warm not hot?* (palatable; also avoids breakdown) *why jaggery not white sugar?* (cultural preference; jaggery is unrefined cane sugar; no major chemistry difference for this purpose).

Either way: 5 minutes of write-up on a reflection slip — “*What did you learn today about claims around turmeric? Pick the most surprising thing.*”

### Wrap-up (5 min)

- Restate the safety rule. Add the turmeric-specific note: **turmeric supplements at multi-gram daily doses have caused liver issues in some individuals, particularly when**

**combined with other supplements.** Culinary use in food is not the same as supplement use.

- Preview Day 5: “Tomorrow, ginger — the kitchen rhizome that’s also a key ingredient in many cold-and-flu home remedies.”

## Student-facing content

**Haridrā** (*Curcuma longa*) — known in English as turmeric, in Hindi as haldi.

**What it is:** A **rhizome** — an underground stem, not a root. Bright orange inside; pale brown outside.

**What’s inside:** **Curcumin** — the polyphenol pigment. Same molecule that gives the yellow colour and most of the studied biological activity.

**The bioavailability problem and the pepper fix:** - Curcumin is poorly absorbed by the gut on its own. - Black pepper (piperine) increases absorption by ~20×. - Traditional Indian cooking often pairs the two — the tradition got there empirically.

**Honest evidence summary:** - Safe and useful as a culinary spice. - Strong lab evidence for anti-inflammatory mechanism. - **Mixed** clinical evidence for specific health claims; **no** evidence supports “cancer cure” claims.

**Traditional Ayurvedic classification:** *kaṭu-tikta* (pungent-bitter) in taste, *uṣṇa* (heating) in potency.

## Homework

- Find turmeric in your home (powder, fresh, capsules, golden milk — anything). Note: where is it? What form? Has anyone in your family ever said “turmeric helps with X”? If yes, write the claim down — we’ll look at it on Day 8.

## Differentiation

- **One tier up:** Curcumin is fluorescent under UV light. Look up *one* other dietary pigment that fluoresces. Why might fluorescence and biological activity be related (hint: think about how light energy interacts with electronic structure)?
- **One tier down:** Focus on identifying the rhizome shape and remembering the word “curcumin.” Stain demo and bioavailability concept come back on Day 10’s revision.

## Teacher notes

- **Stain risk in the classroom is real.** Cover desks. Have warm water + soap nearby. A fresh curcumin stain on cloth washes out before it sets; once it sets and gets sun, it sticks.
- The bioavailability + black pepper story is the **single biggest “aha”** of this module for many students. Their families may have known to use the two together without knowing why. Let students go home and ask.
- Don’t let the “cancer cure” myth go unaddressed. It’s everywhere on social media. Be specific: “lab activity ≠ clinical effect ≠ cure.”

## Citation(s) used in this lesson

- Curcumin chemistry and bioavailability with piperine: well-established in nutrition and ethnopharmacology textbooks. Modern reviews exist in journals such as *Foods*, *Nutrients*, and the *Journal of Ethnopharmacology* — we do not cite a specific paper because RAG verification wasn't available; teachers can search “curcumin piperine bioavailability review” in any indexing database.
- Turmeric as *haridrā* in Ayurveda — standard reference, *Caraka Saṃhitā Sūtrasthāna* (chapter level only).

## Day 5 — *Adraka* (Ginger)

**Module:** Auśadhi · **Band:** Middle · **Time:** 45 min

### Learning objective

By the end of this lesson, students can identify *adraka* (*Zingiber officinale*) as a rhizome, distinguish *adraka* (fresh) from *sūṇṭhī* (dried) and explain that the active compound shifts from **gingerol** to **shogaol** on drying, and describe ginger's traditional classification as *uṣṇa* (heating) with one well-supported modern use.

### Materials

- Fresh ginger rhizome
- Small jar of dried ground ginger powder (*sūṇṭhī*)
- Optional: a thumb-size lump of jaggery for the heating-demo
- Optional, for demo: a small kettle, ginger slices, water, lemon, jaggery — ginger tea
- Whiteboard

### Vocabulary introduced today

- *adraka* (IAST: *adraka*; lit. “ginger”) — fresh ginger rhizome.
- *sūṇṭhī* (IAST: *sūṇṭhī*; lit. “dried ginger”) — same plant, dried rhizome.
- **Gingerol** — the principal pungent compound in fresh ginger ([6]-gingerol).
- **Shogaol** — formed when gingerol loses a water molecule during drying or heating. More pungent. The dominant pungent compound in dried ginger.

### Lesson flow

#### Warm-up (5 min)

Hold up the fresh ginger and the dried ginger powder.

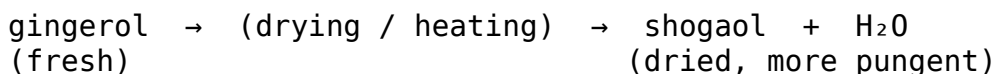
“*Same plant. Same rhizome. One fresh, one dried. Which one will be more pungent?*” Take a vote.

Pass tiny crumb of dried ginger and a tiny shaving of fresh ginger to each student (separate plates). Smell only. Discuss.

## Core (20 min)

1. **Identify it.** Like turmeric, ginger is a **rhizome** — underground stem. The two plants are related (same family: Zingiberaceae). Differences:
  - **Ginger:** pale tan skin; pale yellow-white interior; spicy-pungent smell.
  - **Turmeric:** pale brown skin; bright orange interior; earthy-pepper smell.
  - **Leaves above ground:** very similar — long, blade-like, alternating up a stem.
2. **The fresh-vs-dried switch.** This is the chemistry of the day.
  - **Fresh ginger** is dominated by **[6]-gingerol** — a phenolic compound with a long carbon side-chain ending in a hydroxyl (–OH).
  - When ginger is **dried** (loses water content) or heated, gingerol undergoes a dehydration reaction. The hydroxyl group is lost as water, leaving **[6]-shogaol**.
  - Shogaol is more pungent than gingerol — it's why dried ginger powder is sharper on the tongue than fresh ginger.
  - In Ayurveda, this same transformation is noticed: *adraka* (fresh) is described as somewhat less heating than *sūnṭhī* (dried). The classical text and the modern molecule agree.

Sketch on board:



3. **What ginger does — well-supported uses.**
  - **Anti-nausea** — one of the few traditional herbs with **multiple positive clinical trials**, especially for:
    - Pregnancy-related nausea (under doctor supervision — pregnancy dosing is not a school question).
    - Motion sickness — small ginger doses (1–2 g, e.g. a chewed slice) have outperformed placebo in some trials.
    - Post-surgical nausea — used in some hospital protocols.
  - **Warming feeling** — ginger does cause a measurable rise in skin and body temperature (“thermogenic” effect). Activates TRPV1 receptors — the same receptor activated by capsaicin in chillies.
  - **Digestive support** — culinary use after meals (a slice in tea, or ginger pickle) has a long tradition; mechanism includes stimulation of saliva and gastric secretions.
4. **Traditional Ayurvedic frame.** *Adraka* is *kaṭu* (pungent) in taste, *uṣṇa* (heating) in *vīrya*. Used in:
  - Tea / decoction for cold, congestion, light fever (small culinary amounts).
  - Combined with jaggery for a sore throat (folk remedy; some mechanistic justification through the soothing/warming pairing).
5. **Cautions:**
  - Large doses can cause heartburn or stomach upset.



- Pregnancy — doses higher than culinary need supervision.
- Some blood-thinning medications (warfarin) interact with high ginger intake. Not a school-dose concern but worth knowing.

## Activity (15 min) — Ginger tea demo + Formative Quiz Part B

**First 7 min — Tea demo:** - Teacher slices a 1 cm piece of fresh ginger, drops into 200 ml hot water, lets steep 3–4 minutes. - Optional: add a tiny pinch of jaggery and a drop of lemon. - Pour into a clear cup. Pass around so students can see the colour (very pale yellow) and smell the aroma (sharp, pungent, citrus-warm). - Discussion: which compound dominates here — gingerol or shogaol? (Since the ginger is fresh and only briefly heated: mostly gingerol.) If the ginger had been dried first and then steeped: more shogaol.

**Next 8 min — Formative Quiz Part B** (from quizzes/formative.md). Hand out the 10-question slip. Pencils only. Collect at end of class.

## Wrap-up (5 min)

- Restate safety rule + the anti-nausea note (“ginger has more clinical evidence than most herbs, but for *specific* uses”).
- Preview Day 6: “*Tomorrow we meet a seed that takes the digestive role even further — ajwain.*”

## Student-facing content

**Adraka** (*Zingiber officinale*) — known in English as ginger.

**What it is:** A **rhizome** (underground stem), related to turmeric. Pale tan outside, pale yellow inside.

**Fresh vs dried — same plant, different chemistry:**

| Form          | Sanskrit name | Dominant compound             |
|---------------|---------------|-------------------------------|
| Fresh rhizome | <i>adraka</i> | <b>gingerol</b>               |
| Dried rhizome | <i>sūṇṭhī</i> | <b>shogaol</b> (more pungent) |

When ginger dries or heats, gingerol loses a water molecule and becomes shogaol.

**Well-supported modern uses:** - Anti-nausea, especially motion sickness and pregnancy-related (under medical supervision). - Mild warming effect (activates TRPV1 receptors — same as chilli’s capsaicin). - Digestive support in culinary amounts.

**Ayurvedic classification:** *kaṭu* (pungent) in taste, *uṣṇa* (heating) in *vīrya*.

**Cautions:** Large doses → heartburn or stomach upset; pregnancy doses need a doctor; interactions with blood-thinning medications.

## Homework

- Watch what ginger is used in your home over the next two days. List 3 dishes / drinks that contain it. For each, guess: fresh or dried? Heating or just flavour?

## Differentiation

- **One tier up:** Look up the chemical structure of gingerol and shogaol. Identify the difference (one OH group + one double bond gain). Sketch both in your notebook.
- **One tier down:** Focus on identifying ginger by sight and smell, and remembering “fresh = gingerol, dried = shogaol.”

## Teacher notes

- The chemistry sketch (gingerol → shogaol) is the headline. If a student remembers nothing else from today, they should know that drying ginger changes its chemistry, and the change is well-defined.
- The “ginger has real clinical evidence for nausea” point is important — students often don’t realise that for *some specific herbal uses*, the clinical evidence is genuinely strong. It builds the right kind of nuance: not blanket scepticism, not blanket endorsement, but herb-by-herb, use-by-use evaluation.
- If a student in your class is pregnant (older Middle band, occasionally) or has a pregnant family member raising questions, redirect gently: clinical dosing is for clinicians.

## Citation(s) used in this lesson

- Gingerol → shogaol dehydration on drying: standard pharmacognosy textbook content.
- Ginger as *adraka* (fresh) and *sūṇṭhī* (dried) — distinguished in *Caraka Saṁhitā* and standard Ayurvedic nighantus (lexicons). Chapter-level reference only.
- Anti-nausea clinical evidence: multiple systematic reviews exist; teachers can search “ginger nausea systematic review” in PubMed.

## Day 6 — Yavānī (Ajwain / Carom)

**Module:** Auśadhi · **Band:** Middle · **Time:** 45 min

## Learning objective

By the end of this lesson, students can identify *yavānī* (*Trachyspermum ammi*) by seed appearance and crushed-seed smell, name **thymol** as the principal volatile, and describe one common culinary use linked to traditional digestive support.

## Materials

- Small jar of whole ajwain seeds (one per pair, plus one display jar)
- Pestle and mortar (if available)
- A few sprigs of fresh ajwain leaf (*ajwain ka patta* — sometimes called Indian borage / Cuban oregano — different plant but commonly confused; clarify if you bring it)
- Optional: dried thyme (the European cousin chemistry-wise) for the smell-bridge
- Small bowl of warm water + a teaspoon of ajwain for the demo
- Whiteboard

## Vocabulary introduced today

- *yavānī* (IAST: *yavānī*; lit. “barley-like”) — the Ayurvedic name for ajwain. Also called *ajavāyana* in some texts.
- **Thymol** — the principal volatile aromatic in ajwain seed. Same compound dominates thyme (*Thymus vulgaris*).

## Lesson flow

### Warm-up (5 min)

Hold up the jar of ajwain seeds. Pour a small amount into a tray.

“What do these look like?” (Tiny, oval, brown-grey, ribbed.) “What do you smell?” (Sharp, warm, oregano-like / thyme-like.)

Pass a small pile to each pair. They crush a few seeds with the back of a spoon (no chewing yet) and sniff. The smell intensifies dramatically when crushed — release of volatile thymol.

### Core (20 min)

#### 1. Identify it.

- *Yavānī* is a small annual herb in the Apiaceae family (same family as coriander, cumin, fennel, dill).
- The “seeds” sold in shops are technically **fruits** (small schizocarps), about 2 mm long, oval, ridged, greyish-brown.
- Crushed, they release a strong thyme-like aroma. The smell is the chemistry — thymol is volatile, so crushing exposes more of it.

#### 2. The ajwain-leaf confusion (1 min).

What’s sold in Indian markets as “ajwain leaf” or “ajwain ka patta” is often *Plectranthus amboinicus* — Cuban oregano. It’s a different plant, but it smells similar (both contain thymol-class compounds). Modern botany distinguishes them; in many kitchens they get used somewhat interchangeably. Note for clarity but don’t dwell.

#### 3. What’s inside.

Thymol is a **phenolic monoterpenoid**,  $C_{10}H_{14}O$  — closely related to carvacrol. Properties:

- **Antimicrobial** in vitro — used historically in mouthwashes and antiseptic preparations (Listerine famously included thymol).
- **Carminative** — helps reduce intestinal gas. Mechanism likely involves smooth-muscle relaxation in the gut.
- **Strong volatile** — that’s why a tightly closed spice jar holds the smell, and an open one fades in a few weeks.

#### 4. Traditional use — culinary, not medicinal.

- Tempered into hot oil at the start of cooking parathas, kachoris, savoury snacks.

- **Ajwain water** — a common Indian household remedy: 1 tsp ajwain steeped in 200 ml hot water for 5–10 minutes, strained, drunk warm. Used for indigestion, bloating, mild stomach discomfort.
- Often added to flour-based foods that are heavy on starch (parathas, samosas) — the carminative effect helps with the digestive load.

#### 5. **Honest evidence summary.**

- **Mechanism evidence:** thymol relaxes gut smooth muscle and has antimicrobial action. Well established at the lab level.
- **Clinical evidence:** for *general indigestion / dyspepsia*, the clinical trial picture is small but generally positive for thymol-containing preparations.
- **Cautions:** essential oils made from ajwain are extremely concentrated — multi-fold more thymol per drop than the whole spice. **Not for ingestion at school age.** The whole seed in food is a different scale entirely.

#### 6. **The doṣa frame.** *Yavānī* is *kaṭu* (pungent) and *tikta* (bitter), *uṣṇa* (heating). Considered to balance *vāta* and *kapha* (mainly by stimulating digestion).

### Activity (15 min) — Ajwain-water demo + Compare to thyme

**Demo (5 min):** - Teacher places 1 tsp ajwain in a clear glass. Pours 200 ml hot (not boiling) water over it. Covers with a saucer. - Pass around the glass — observe the seeds float, the water gain colour, the smell rise. - 5 minutes later, uncover. The aroma is now intense. - This is the home remedy: strain and drink warm (for the demo, don't drink — just observe).

**Compare to thyme (5 min):** - Pass around dried thyme. Crush a small amount, smell. - The two herbs are unrelated botanically (different families) but share the same dominant volatile — **thymol**. This is a beautiful example of *convergent chemistry* — different plants evolving the same compound because it serves a similar purpose for the plant (defence against microbes and herbivores).

**Exit ticket (5 min):** - On a slip: “Name today's herb in Sanskrit, its main compound, and one common Indian dish you've eaten it in.”

### Wrap-up (5 min)

- Restate safety rule + the essential-oil warning (“the whole seed in food is fine; the essential oil is concentrated and not for school-age ingestion”).
- Preview Day 7: “Tomorrow — mint. The cooling herb.”

### Student-facing content

***Yavānī*** (*Trachyspermum ammi*) — known in English as ajwain or carom seed.

**What it is:** A small annual herb in the same family as coriander, cumin, and fennel (Apiaceae). The “seeds” you buy are technically tiny fruits.

**How to identify:** Oval, ridged, greyish-brown, about 2 mm long. Strong thyme-like smell — much stronger when crushed.

**What's inside: Thymol** — a phenolic monoterpenoid. Same dominant compound as European thyme — *convergent chemistry* between unrelated plants.

**Traditional use:** Tempered into hot oil for parathas, savoury snacks. Ajwain water (steeped seeds in warm water) is a common Indian household remedy for indigestion.

**Honest evidence:** Mechanism for carminative (anti-gas) action is well established. Clinical evidence for general indigestion is positive but small. **Essential oils are concentrated and not safe for ingestion in school-age use.**

**Ayurvedic frame:** *kaṭu-tikta* (pungent-bitter), *uṣṇa* (heating). Balances *vāta* and *kapha* via digestive stimulation.

## Homework

- Find ajwain in your kitchen. Smell it. Then ask one adult: “*What do you put ajwain in, and why?*” Write down their answer.

## Differentiation

- One tier up:** *Convergent chemistry* — find one other example of two unrelated plant families producing the same compound (e.g. cinnamaldehyde in cinnamon and in some flowers; geraniol in geranium, lemongrass, rose). Sketch the molecular structure of thymol if you can.
- One tier down:** Focus on identifying ajwain seeds by smell and naming the compound “thymol.”

## Teacher notes

- The **crushed-seed smell intensification** is the most memorable hands-on of the day. Make sure every student gets to crush their own seeds. The “oh wow” reaction is the lesson.
- The Indian-borage / ajwain-leaf confusion is real and you’ll get questions. Acknowledge — “the leaf you find in shops is from a different plant that smells similar” — but don’t make the module about taxonomy.
- The ajwain-water demo is one of the safer cooking-adjacent demos. Even so: don’t have students drink from a shared glass. Demonstrate; don’t distribute.

## Citation(s) used in this lesson

- Thymol as principal volatile in *Trachyspermum ammi*: standard ethnopharmacology / phytochemistry reference.
- Carminative mechanism of thymol: well-established at lab level; clinical evidence reviewed in standard ethnopharmacology journals. No specific paper cited.
- Yavānī* in Ayurveda — standard nighantu references; chapter-level only.

## Day 7 — *Pudīnā* (Mint)

**Module:** Auśadhi · **Band:** Middle · **Time:** 45 min

## Learning objective

By the end of this lesson, students can identify *pudīnā* (*Mentha* species) by leaf shape and the characteristic cooling smell, name **menthol** as the principal volatile, distinguish two common varieties (spearmint vs peppermint), and explain that mint is classified as *śīta* (cooling) in Ayurveda — a classification that aligns with its measurable physiological effect.

## Materials

- Two sprigs of mint, ideally from two different varieties (spearmint and peppermint are easiest to source)
- A small jar of dried mint leaves
- A few peppermint candies or a small tube of menthol-containing balm — for the smell bridge (do not let students apply balm to their skin in class)
- Ice cube + a small bowl (for the demo on cooling sensation)
- Whiteboard

## Vocabulary introduced today

- *pudīnā* (IAST: pudīnā) — mint. Commonly refers to *Mentha spicata* (spearmint) in Indian kitchens.
- **Menthol** — the principal volatile cooling compound. Most concentrated in peppermint (*Mentha × piperita*); also present in smaller amounts in spearmint.
- *śīta* (IAST: śīta; lit. “cool, cooling”) — the cooling potency (*vīrya*) in Ayurveda. Contrasts with *uṣṇa* (heating).

## Lesson flow

### Warm-up (5 min)

Pass the mint sprigs along the row. Each student takes a leaf, rubs gently, sniffs.

“Three words for the smell.” (Likely: “fresh,” “cool,” “toothpaste-like.”)

Hold an ice cube briefly — note how cold feels on the skin. Now ask: “When you smelled the mint just now, did it feel like the ice or different?” (Different — but related. We’ll explain.)

### Core (20 min)

#### 1. Identify it.

- **Mint family (Lamiaceae)** — same family as tulsi (square stems!). Make students check: press the mint stem between thumb and finger. Four flat sides.
- **Leaf shape:** oval-pointed, serrate (jagged) edges. Larger than tulsi leaves. Bright green.
- **Smell:** the unmistakable cooling-fresh aroma. Different varieties smell slightly different.
- **Common varieties in India:**
  - *Mentha spicata* — spearmint. The everyday “pudina” in chutney, raita.

- *Mentha × piperita* — peppermint. More concentrated menthol; sharper, “tooth-paste” smell.
- *Mentha arvensis* — corn mint / Japanese mint — grown commercially in India for menthol extraction.

## 2. **The cooling trick — explained.** This is the headline.

- Menthol activates a receptor in your skin called **TRPM8** — the same receptor that activates when your skin actually gets cold.
- The receptor doesn’t know whether the signal is from real cold or from a menthol molecule. It sends “cold!” to your brain either way.
- The result: **you feel cool without your temperature actually dropping.** Your skin temperature is unchanged. Your perception of coolness is real.
- This is the same trick that capsaicin (chilli) plays in reverse — capsaicin activates TRPV1, the “heat” receptor, and your brain feels heat even though your skin isn’t hot.

This is one of the cleanest connections between **traditional classification** (*śīta* — cooling) and **modern molecular pharmacology** (TRPM8 activation). The tradition got the *effect* right; modern science explained the *mechanism*.

## 3. **What mint does — well-supported.**

- **Cooling sensation** (TRPM8) — definitively established.
- **Digestive support** — peppermint oil capsules have **good clinical evidence** for relieving symptoms of Irritable Bowel Syndrome (IBS) when prescribed by a doctor. Mechanism: relaxation of gut smooth muscle.
- **Breath freshness** — antibacterial activity in the mouth.
- **Headache relief** when applied to the temples (the cooling sensation distracts from pain). Topical use of dilute menthol; not the same as ingesting essential oil.

## 4. **Cautions.**

- **Babies and small children** — undiluted menthol-containing products on the chest or face can cause breathing reflexes. Do not give menthol balms to infants.
- **Peppermint oil capsules** — these are *medications*, not snacks. The IBS evidence is for *prescribed* dosing.
- **Acid reflux** — peppermint can relax the lower oesophageal sphincter, making reflux worse for some people.

## 5. **Traditional Ayurvedic frame.** *Pudīnā* is *kaṭu* (pungent) in taste but *śīta* (cooling) in *vīrya*. Used in summer drinks (mint nimbu pani), in chutneys (with coriander and lime), and in some preparations for cooling *pitta* imbalances.

## Activity (15 min) — Variety taste-id challenge

(Skip if cooking/tasting is not permitted at your school. Replace with a sketch-by-variety exercise.)

- Set up two cups: spearmint chutney (from prepared chutney, very small portion per student) vs a peppermint candy crushed in water (tiny amount).
- Each pair tries to identify which is which by taste alone.
- Record what made them distinguishable — sharpness? Sweetness? Cooling intensity?

**Final 5 min — group discussion:** - “*Why does mint feel cool when it isn’t actually cold?*” (TRPM8 activation — students should be able to explain this back.) - “*What’s the connection between the Ayurvedic word *śīta* and this molecular trick?*” (The tradition was tracking the effect; the science explains the cause.)

### Wrap-up (5 min)

- Restate the safety rule.
- Preview Day 8: “*Tomorrow we put all six herbs together — and look at them through the *doṣa* lens. Heating, cooling, what goes with what.*”

## Student-facing content

**Pudīnā** (*Mentha* species) — mint.

**What it is:** A Lamiaceae herb (square stems!). Common varieties in India: *Mentha spicata* (spearmint), *Mentha × piperita* (peppermint), *Mentha arvensis* (corn mint).

**How to identify:** Oval-pointed serrate leaves, bright green; the unmistakable cooling-fresh smell when crushed.

**What’s inside: Menthol** — the principal volatile.

**Why does mint feel cool?** - Menthol activates **TRPM8** — the same receptor in your skin that activates when you actually feel cold. - Your brain receives the “cold!” signal even though your skin’s temperature hasn’t changed. - This is the **mechanism** behind the Ayurvedic classification of mint as *śīta* (cooling) — the tradition observed the effect; modern pharmacology explained the cause.

**Well-supported modern uses:** - Cooling sensation (mechanism: TRPM8 activation). - Peppermint oil for IBS symptoms (prescribed, not self-administered). - Antibacterial action in the mouth (breath freshness).

**Cautions:** Not for infants on the chest/face. Concentrated essential oil and capsules are medications, not snacks.

## Homework

- Find pudina (or any mint) in your home or near it. Sketch the leaf. Identify which variety if you can. Note: where do you encounter mint most often in your week?

## Differentiation

- **One tier up:** TRPM8 was identified as the cold receptor in 2002. Look up the David Julius lab work on TRP channels (2021 Nobel Prize in Physiology or Medicine). Be ready to share one finding next class.
- **One tier down:** Focus on identifying mint by smell and remembering “menthol = cool feeling.”



## Teacher notes

- The **TRPM8 mechanism** is the single most rewarding teaching moment of this module for a science-leaning student. The bridge from “cooling herb” (tradition) to “TRPM8 activation” (Nobel-winning science) is satisfying and intellectually honest.
- If anyone in your class connects “menthol cooling = chilli burning = same kind of receptor trick,” that’s the inferential leap to highlight publicly. Praise it.
- A few students may bring up vapour rubs / chest balms used in childhood. Acknowledge the use; note the infant caution; don’t sermonise.

## Citation(s) used in this lesson

- Menthol activation of TRPM8: established in molecular pharmacology. David Julius and colleagues identified TRPM8 as the cold/menthol receptor (Julius lab, UCSF; 2021 Nobel Prize work). Reference is at the lab level only; specific paper not cited.
- Peppermint oil for IBS: multiple systematic reviews exist. Teachers can search “peppermint oil IBS systematic review” in PubMed.
- *Pudīnā* as *śīta* — standard Ayurvedic nighantu reference; chapter-level only.

# Day 8 — The Doṣa Lens: Heating and Cooling

**Module:** Auśadhi · **Band:** Middle · **Time:** 45 min

## Learning objective

By the end of this lesson, students can sort all six module herbs into heating (*uṣṇa*) and cooling (*śīta*) categories, explain the Ayurvedic concept of *vīrya* in one sentence, link this classification to specific *doṣa* imbalances from Module 4, and identify one example where the traditional classification matches a measurable modern pharmacological effect.

## Materials

- A printed 6-card set with each herb (one per pair); these can be drafts of the project deck
- Whiteboard with two columns: **uṣṇa** (heating) | **śīta** (cooling)
- Sticky notes
- Optional: Day 4–7 worksheets students completed, so they have their own data

## Vocabulary introduced today

- *vīrya* (IAST: *vīrya*; lit. “potency, energy”) — in Ayurveda, the heating or cooling *potency* of a substance, distinct from its taste (*rasa*) and its post-digestive effect (*vipāka*). At Middle band we use only the heating/cooling axis.
- *uṣṇa* (IAST: *uṣṇa*; lit. “hot, heating”) — the heating *vīrya*.
- *śīta* (IAST: *śīta*; lit. “cold, cooling”) — the cooling *vīrya*.

## Lesson flow

### Warm-up (5 min)

Daily prompt: “Which herb did you smell or eat at home since our last class?” Two students share for 30 sec each.

Now the day’s hook: “From what you remember about each herb — which feels ‘warming’ when you eat it? Which feels ‘cooling’?” Take a few intuitions, write them on the board.

### Core (20 min)

1. **What is *vīrya*?** Write on board:

*vīrya* — the **heating** or **cooling** potency of a substance, measured by its effect on the body.

This is one of four properties Ayurveda uses to classify any food or herb:

- *rasa* — taste (sweet, sour, salty, pungent, bitter, astringent)
- *vīrya* — heating or cooling potency
- *vipāka* — post-digestive effect
- *prabhāva* — special, idiosyncratic action

At Middle band, we focus on *vīrya*. The full fourfold framework comes back at Senior band.

2. **Sort the six herbs.** Build the two-column table on the board, asking students to predict before you confirm:

| <i>uṣṇa</i> (heating)                                | <i>śīta</i> (cooling)                  |
|------------------------------------------------------|----------------------------------------|
| <i>tulasī</i> — eugenol, warming aromatic            | <i>pudīnā</i> — menthol, TRPM8 cooling |
| <i>haridrā</i> — turmeric, mild warming              |                                        |
| <i>adraka</i> — ginger, strongly warming (TRPV1)     |                                        |
| <i>yavānī</i> — ajwain, warming digestive            |                                        |
| <i>nimba</i> — neem (slightly contested — see below) |                                        |

**The contested case:** *nimba* (neem) is classified as bitter (*tikta*) and *śīta* (cooling) in classical Ayurveda — even though it’s a bitter herb without an obviously cooling sensation. Why? Because in the *doṣa* model, bitterness reduces *pitta* (the “fire” *doṣa*) — and a substance that reduces *pitta* is by definition cooling in the system. Bring this up explicitly to show students that the Ayurvedic categories are **functional**, not always sensory.

3. **Why does heating/cooling matter?** Link back to *doṣas* (refresh briefly if Module 4 hasn’t been taught):

- **Vāta** (motion, air-and-space) — generally treated with warming, moistening, grounding foods/herbs. Cold and dry aggravate *vāta*.

- **Pitta** (transformation, fire-and-water) — treated with cooling, mild foods. Heat and pungency aggravate pitta.
- **Kapha** (structure, water-and-earth) — treated with warming, light, drying foods. Cold and heavy aggravate kapha.

So:

- **Ginger + ajwain** → useful for *kapha* and *vāta* imbalances (warming, stimulating).
  - **Mint** → useful for *pitta* imbalances (cooling).
  - **Tulsi** → balanced for *vāta*/*kapha* (warming, but not as strongly as ginger).
  - **Turmeric** → mildly heating but broadly used because of its antimicrobial profile.
  - **Neem** → useful for *pitta* and *kapha* imbalances (its bitterness reduces both, in the framework’s logic).
4. **The match between tradition and modern pharmacology.** This is the most important slide of the module:

| Herb                      | Tradition says        | Modern says                                       |
|---------------------------|-----------------------|---------------------------------------------------|
| <i>pudīnā</i><br>(mint)   | <i>śīta</i> — cooling | Menthol activates TRPM8 → “cool” signal to brain  |
| <i>adraka</i><br>(ginger) | <i>uṣṇa</i> — heating | Gingerol activates TRPV1 → “warm” signal to brain |

Both classifications were made by careful observers — millennia apart, using completely different methods. They agree because they were both tracking a real biological effect.

5. **Source — paraphrase of the Auṣadhi-sūkta** (Ṛgveda 10.97). Read aloud once:

*“O herbs, born before the gods, brown and bright, born of past ages — may you who carry many forms protect the one who calls on you.”*

— paraphrase of the opening of Ṛgveda 10.97, the *Auṣadhi-sūkta*

Comment: the hymn is **3000+ years old**. The plants are addressed as agents — like persons. Whether or not you take that literally, notice what it does: it treats the plant as something to **pay attention to**. The *doṣa* framework is built on millennia of paying that kind of attention.

**Activity (15 min) — Sort by vīrya** (see [activities/activity-02-sort-by-virya.md](#))

Pairs receive a set of 12 cards: the 6 module herbs + 6 other common kitchen items (rice, ghee, lemon, ice cream, chilli, coriander seed). They sort into *uṣṇa* / *śīta* / *not sure* and defend their reasoning.

Last 3 min: each pair shares their **most uncertain** card and why.

**Wrap-up (5 min)**

- Restate safety rule.

- Hand out the project deck instructions for Day 9. Show the rubric on the board. Walk through one card's expected content.
- Preview Day 9: “*Tomorrow you build your six-card herb deck. Bring your sketches, your notes, your colour pencils.*”

## Student-facing content

**Vīrya** is the heating or cooling *potency* of a substance in Ayurveda.

**Heating (*uṣṇa*)** — tulsī, turmeric, ginger, ajwain. **Cooling (*śīta*)** — mint. Neem too — **though in a functional, not sensory, way.**

**Why does this matter?** - Heating herbs balance *vāta* and *kapha* imbalances (cold-related, sluggish conditions). - Cooling herbs balance *pitta* imbalances (heat-related, inflammatory conditions).

**Two beautiful matches between tradition and modern science:** - *Pudīnā* is *śīta* — menthol activates TRPM8, the same skin receptor that fires when you actually feel cold. - *Adraka* is *uṣṇa* — gingerol activates TRPV1, the same skin receptor that fires when you actually feel heat.

Both classifications were made by careful observers, millennia apart, using completely different methods. They agree because they were tracking a real biological effect.

“O herbs, born before the gods, brown and bright, born of past ages — may you who carry many forms protect the one who calls on you.” — paraphrase of the opening of the *Auṣadhi-sūkta* (R̥gveda 10.97)

## Homework

- Finalise your project pair. Decide which 2 herbs you'll each draft cards for first (you'll do 3 each). Bring everything tomorrow.
- One sentence answer for the next class: “*What is the most surprising thing you've learned about herbs in the past week?*”

## Differentiation

- **One tier up:** The full Ayurvedic fourfold framework is *rasa-vīrya-vipāka-prabhāva*. Look up what *vipāka* means and find one example where a herb's *vipāka* differs from its *rasa*.
- **One tier down:** Focus on the heating/cooling sort. The *doṣa* connection comes back in summative review.

## Teacher notes

- The TRPM8/TRPV1 ↔ *śīta/uṣṇa* convergence is the cleanest “tradition and modern science describe the same biological reality” moment in the module. Lean into it. Don't oversell (“Ayurveda discovered TRP channels first” — no, they didn't, that's a different question) — but don't undersell either.
- The neem-is-cooling-because-bitter point will surprise students. Use it to discuss what “potency” means in a functional framework vs a sensory one.

- The Auṣadhi-sūkta paraphrase is *not* a translation. Don't pass it off as a quote. Be explicit: this is a paraphrase of the opening of Ṛgveda 10.97. Teachers who want a published translation should consult Griffith (1896) or Jamison & Brereton (2014).

### Citation(s) used in this lesson

- *Auṣadhi-sūkta* — Ṛgveda 10.97 — referenced and paraphrased. Public-domain Vedic source.
- *Caraka Saṁhitā* — *vīrya* as one of four properties of substances. Chapter-level reference to *Sūtrasthāna*.
- TRPM8 / TRPV1 work — established neuroscience. 2021 Nobel Prize in Physiology or Medicine to David Julius and Ardem Patapoutian.

## Day 9 — Project Session: Build the Herb Deck

**Module:** Auṣadhi · **Band:** Middle · **Time:** 45 min

### Learning objective

By the end of this lesson, each pair has produced **all six cards** in draft form, each containing the eight required elements (see below). The cards are ready for finalising at home and presenting on Day 10.

### Materials

- Pre-cut A6 (or quarter-A4) cards — 6 per pair, plus 2 spares
- Colour pencils, fine-tip markers, scale rulers
- Reference: students' Day 2–8 worksheets (instruct them to bring these)
- Project brief ([assessments/project-brief.md](#)) printed and visible
- Rubric ([assessments/rubric.md](#)) printed and visible

### Lesson flow

#### Warm-up (5 min)

Read the project brief aloud (one student volunteers).

Restate the **eight required elements** on each card: 1. Sketch of the herb (leaf, root, or seed — whichever is most identifying) 2. **Sanskrit name** in IAST 3. **English / common name** 4. **Botanical name** (italicised) 5. **Family** (e.g. Lamiaceae, Apiaceae, Zingiberaceae) 6. **Three identifying features** (smell, leaf shape, colour, etc.) 7. **Active compound** (curcumin, eugenol, etc.) 8. **One traditional use + one safety note**

Show one finished card example (you can use yesterday's *pudīnā* card-template).

#### Core / activity (35 min)

Pairs work continuously. The teacher rotates and intervenes.

### Pacing checkpoints — write on the board:

- **By minute 12 (out of 35):** All six cards should have the sketch + Sanskrit + English name. (The fastest part.)
- **By minute 25:** All six cards should have the active compound and one traditional use.
- **By minute 35:** Each card should have the safety note + three identifying features.

If a pair lags at any checkpoint, intervene specifically — don't let them spend 15 minutes perfecting one card.

**Quality interventions to watch for:** - Cards copying directly from a worksheet without engagement → ask “what's the most interesting thing about this herb? Add that to the card.” - Cards with no safety note → “what's the dose question for this herb?” Write the answer. - Cards with grand health claims (“turmeric cures arthritis”) → push back: “what does the evidence actually say?” - Cards missing the family → look it up together on the board (Lamiaceae for tulsi+mint, Apiaceae for ajwain, Zingiberaceae for ginger+turmeric, Meliaceae for neem).

### Wrap-up (5 min)

- Each pair holds up one card they're proudest of. 10-second show + tell.
- Collect cards in pair-folders for tomorrow. (Or have pairs take them home if they prefer to finalise overnight.)
- Hand out the reflection slip — to complete tonight as homework: > “*What I set out to learn, what I actually learned, and one question I still have.*”

### Student-facing content

**Today: build your 6-card herb deck.**

Each card has eight things — sketch, Sanskrit name, English name, botanical name, family, three identifying features, active compound, and one traditional use + one safety note.

Quality over completeness — but all six cards must be drafted by end of class.

Use what you've already drawn and noted across Days 2–7. Your worksheets are your raw material; the cards are the polished version.

### Homework

- Finalise any card you didn't complete in class.
- Complete the reflection slip.
- Bring all six cards + reflection slip + pencils to Day 10.

### Differentiation

- **One tier up:** Make a **7th card** for a herb of your choice — *aśvagandhā*, *brāhmī*, *triphālā*, *kariveppilai* (curry leaf), or *dhānyaka* (coriander). Apply the same eight-element template.
- **One tier down:** Make 4 cards (drop turmeric and ajwain if compressed) — but each must have all 8 elements. Quality over quantity.

## Teacher notes

- This is a build session, not a teaching session. Resist the urge to lecture. Walk around. Ask questions. Point at gaps.
- Common failure mode: pairs spend the first 15 min discussing what to draw and end up with one half-card. Set the timer visible. Push pace.
- Common failure mode: one partner draws while the other watches. Reassign — partner A does 3 cards in pen, partner B does 3 cards in pen, then they swap and refine each other's.

## Citation(s) used in this lesson

- Project brief — [assessments/project-brief.md](#)
- Rubric — [assessments/rubric.md](#)

# Day 10 — Showcase + Summative Quiz

**Module:** Auṣadhi · **Band:** Middle · **Time:** 45 min

## Learning objective

Students present their herb deck to the class (2 minutes per pair), complete the summative quiz (20 min, 25 marks), and consolidate the module's safety message and tradition-meets-science framing.

## Materials

- All pairs' completed decks
- Printed copies of [quizzes/summative.md](#) — one per student
- Rubric ([assessments/rubric.md](#)) for in-the-moment marking
- Timer
- Whiteboard

## Lesson flow

### Warm-up (3 min)

- Restate the safety rule one last time, chorally.
- Quick housekeeping: pairs sit together, decks in front of them, name and roll number on a slip clipped to the deck.

### Summative quiz (20 min)

- Distribute the quiz papers. Pencils only. Closed book.
- Strict timing: 20 minutes.
- Collect at minute 20.

### Project presentations (20 min)

- 2 minutes per pair (10 pairs ≈ full class; if more, adjust to 1:30 each).

- Each pair picks **one card** from their deck to present in detail. The other five are passed around or laid out for inspection.
- Mark rubrics in real-time. Don't try to mark after class — the energy fades.
- Q&A after each presentation is **30 seconds**: one classmate's question, one pair's answer. Move on.

### Wrap-up (2 min)

- Collect decks for archiving (or return to pairs if they want them home — they earned them).
- One closing reflection — go around the room, each student in 5 words: *“What stays with you?”*
- Close: *“Three things to remember from this module: food amounts are safe; tradition and science can describe the same biological reality; ask ‘how much’ before you ask anything else about an herb.”*

### Student-facing content

**Today is showcase + quiz.**

1. Closed-book quiz, 20 min, 25 marks.
2. Present your favourite card from the deck for 2 minutes. Be ready for one quick question.
3. Leave with: the safety rule in your head, your deck in your hand, one fact you didn't know two weeks ago.

### Differentiation

- **Tier up:** Volunteers can present **for** 3 minutes by request, presenting two cards — one of the six standard and one bonus card they made.
- **Tier down:** A student who freezes can have their pair-partner present while they hold up the card. Don't make this a failure mode.

### Teacher notes

- The mark on the rubric **as the pair presents** — not after. Trust your first read.
- If the quiz takes longer than 20 minutes for the slowest students, extend by 5 max. Don't let it eat into presentations.
- Reflection question at the end (“what stays with you?”) is the most honest indicator of what landed. Note it down if you can — useful for adjusting Year 2.

### Citation(s) used in this lesson

- Quiz: `quizzes/summative.md`
- Rubric: `assessments/rubric.md`
- Project brief: `assessments/project-brief.md`



## Module 10 (Middle) — Formative Quizzes

Two parts: Part A on Day 3 (5 questions, 5 min); Part B on Day 5 (10 questions, 10 min). Both are diagnostic — not graded toward the final mark. Hand back within 24 hours.

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### Part A — Day 3 (5 min, 5 questions)

Name: \_\_\_\_\_ Class: \_\_\_\_\_

1. What does *auṣadhi* mean?

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2. Tulsi (*Ocimum tenuiflorum*) belongs to which plant family?

- a) Apiaceae (parsley family)
- b) Lamiaceae (mint family)
- c) Zingiberaceae (ginger family)
- d) Meliaceae (mahogany family)

3. The principal aromatic compound in tulsi is also the dominant compound in clove. What is it called?

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4. True or false: A neem leaf in concentrated form is safe to eat in any quantity. ( T / F )

5. Why does crushing a leaf release more smell than just looking at it?

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### Part A — answer key

| Q | Answer                                                                                                                 | Notes                                                    |
|---|------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| 1 | “Herb” / “plant used for healing” / “medicinal plant”                                                                  | Any phrasing close to this.                              |
| 2 | b) Lamiaceae                                                                                                           | The square-stem family.                                  |
| 3 | eugenol                                                                                                                | Spelling tolerated if recognisable.                      |
| 4 | F                                                                                                                      | Concentrated neem (especially oil) is hepatotoxic.       |
| 5 | The aromatic compounds are <i>volatile</i> — crushing breaks plant cell walls and releases the compounds into the air. | Any answer mentioning volatiles, oils, or cell breakage. |

## Part B — Day 5 (10 min, 10 questions, mixed format)

Name: \_\_\_\_\_ Class: \_\_\_\_\_

1. (1 mark) Write the Sanskrit name for ginger.

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2. (1 mark) Write the Sanskrit name for turmeric.

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3. (1 mark) Both ginger and turmeric are **rhizomes**, not roots. What is the difference between a rhizome and a root?

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4. (1 mark) The orange colour of turmeric comes from one specific molecule. What is it called?

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5. (2 marks) Black pepper (piperine) increases the absorption of curcumin by roughly 20×. What does this tell us about (a) why turmeric is traditionally cooked with pepper, and (b) why a turmeric supplement without pepper might not work as advertised?

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6. (1 mark) Fresh ginger contains mainly \_\_\_\_\_. Dried ginger contains mainly \_\_\_\_\_.

7. (1 mark) True or false: The fresh-to-dry change in ginger's chemistry involves the loss of a water molecule. ( T / F )

8. (1 mark) Match each herb to its plant family:

|        |                               |
|--------|-------------------------------|
| tulasī | Apiaceae (parsley family)     |
| nimba  | Lamiaceae (mint family)       |
| yavānī | Zingiberaceae (ginger family) |
| adraka | Meliaceae (mahogany family)   |

9. (1 mark) Name one well-supported modern clinical use of ginger.

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## Part B — answer key

| Q | Answer                                    | Mark scheme |
|---|-------------------------------------------|-------------|
| 1 | <i>adraka</i> (accept also <i>adrak</i> ) | 1           |

| Q | Answer                                                                                                                                                | Mark scheme                                         |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| 2 | <i>haridrā</i> (accept also <i>haldi</i> )                                                                                                            | 1                                                   |
| 3 | A rhizome is an underground stem that stores starch and produces shoots; a root absorbs water and nutrients.                                          | 1 if mentions “stem vs root” or “underground stem.” |
| 4 | curcumin                                                                                                                                              | 1; spelling tolerated.                              |
| 5 | (a) Tradition empirically discovered the synergy. (b) Curcumin alone is poorly absorbed; without piperine (or fat/lecithin), it’s largely unabsorbed. | 2 if both points; 1 if either.                      |
| 6 | gingerol; shogaol                                                                                                                                     | 1 if both correct; 0 otherwise.                     |
| 7 | T                                                                                                                                                     | 1                                                   |
| 8 | tulasī ↔ Lamiaceae,<br>nimba ↔ Meliaceae,<br>yavānī ↔ Apiaceae,<br>adraka ↔ Zingiberaceae                                                             | 1 if all four; 0.5 if 2–3 correct.                  |
| 9 | Anti-nausea (motion sickness, post-surgical, or pregnancy-related under supervision).                                                                 | 1 if any specific clinical use named accurately.    |

Total Part B: 10 marks. Not added to summative.

## What to do with the results

- A class average below 6/10 on Part B = significant gaps. Reteach the chemistry headlines on Day 6 (push half of the ajwain content to Day 7).
- Individual students below 3/10 on Part B = pull aside on Day 7 for a 10-min one-on-one before the activity starts.

## Module 10 (Middle) — Summative Quiz

**Time:** 20 min · **Marks:** 25 · **Closed book**

**Instructions:** Write your name and class. Answer all questions. Section D has one mark for clear handwriting and sentence structure. No calculators needed.

**Name:** \_\_\_\_\_ **Class:** \_\_\_\_\_

## Section A — Multiple choice ( $5 \times 1 = 5$ marks)

1. Which of the following is the principal aromatic compound in tulsi?
    - a) curcumin
    - b) eugenol
    - c) menthol
    - d) thymol
  2. Both turmeric and ginger are:
    - a) roots
    - b) stems above ground
    - c) rhizomes (underground stems)
    - d) bulbs
  3. Why does mint feel cooling on the tongue even though it doesn't lower the temperature?
    - a) Menthol cools the air around it.
    - b) Menthol activates a temperature-sensing receptor (TRPM8) that normally responds to cold.
    - c) Mint contains ice crystals.
    - d) Mint reduces blood flow to the tongue.
  4. Which herb is classified as *tikta* (bitter) and is well known for its azadirachtin content?
    - a) tulasī
    - b) haridrā
    - c) nimba
    - d) yavānī
  5. "Food amounts are safe; therapeutic doses are a clinical question." This sentence is most accurately a statement about:
    - a) why herbs don't work
    - b) the difference between culinary use and concentrated medicinal use
    - c) modern science being superior to Ayurveda
    - d) the doṣa framework
- 

## Section B — Short answer ( $4 \times 2 = 8$ marks)

6. Name the principal active compound for each of the following herbs:

| Herb                          | Compound |
|-------------------------------|----------|
| <i>haridrā</i> (turmeric)     | _____    |
| <i>adraka</i> (ginger, fresh) | _____    |

| Herb                   | Compound |
|------------------------|----------|
| <i>yavānī</i> (ajwain) | _____    |
| <i>pudīnā</i> (mint)   | _____    |

7. When fresh ginger is dried, gingerol is converted into a different compound that is more pungent. Name the new compound. What atom or group does the molecule lose in this conversion?

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8. *Vīrya* in Ayurveda refers to a substance's heating or cooling potency. Sort the following herbs into the two categories:

tulasī, nimba, haridrā, adraka, yavānī, pudīnā

|                       |                       |
|-----------------------|-----------------------|
| <i>Uṣṇa</i> (heating) | <i>Śīta</i> (cooling) |
|-----------------------|-----------------------|

9. Why is concentrated neem (especially neem oil) not safe for ingestion? Name one specific organ that can be affected.

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## Section C — Identify and explain (1 × 5 = 5 marks)

10. Below is a “found at the kitchen counter” description of a plant or part. For each, identify the herb and one identifying feature.

- (a) “A small ridged seed-like fruit, about 2 mm long, greyish-brown, with a strong thyme-like smell when crushed.”

Herb: \_\_\_\_\_

Feature: \_\_\_\_\_

- (b) “A fingerlike underground stem with pale brown skin and bright orange interior; permanent yellow stain on the cutting board.”

Herb: \_\_\_\_\_

Feature: \_\_\_\_\_

(c) “A small green leaf with toothed edges on a square stem, sharp clove-like smell when crushed.”

Herb: \_\_\_\_\_

Feature: \_\_\_\_\_

(d) “A compound leaf with 9–15 sickle-shaped leaflets along a central stem; intensely bitter.”

Herb: \_\_\_\_\_

Feature: \_\_\_\_\_

(e) “A bright green leaf with serrate edges on a square stem, distinctive cooling smell.”

Herb: \_\_\_\_\_

Feature: \_\_\_\_\_

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## Section D — Open response (1 × 7 = 7 marks; 1 of which is for clarity)

**11.** You’re scrolling social media and see a post claiming: “*Turmeric tea every morning will boost your immunity and prevent cancer.*”

In 5–7 sentences: - Identify what part of this claim is reasonable. - Identify what part is overclaimed. - Explain the bioavailability problem with curcumin and what tradition does to address it. - Conclude with what would be a more honest version of the claim.

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## Answer key (teacher copy)

| Q | Answer                     | Mark scheme |
|---|----------------------------|-------------|
| 1 | b) eugenol                 | 1           |
| 2 | c) rhizomes                | 1           |
| 3 | b) Menthol activates TRPM8 | 1           |

| Q  | Answer                                                                                                                                                                                                                                                                                                | Mark scheme                                                                                                                                                          |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4  | c) nimba                                                                                                                                                                                                                                                                                              | 1                                                                                                                                                                    |
| 5  | b) the difference between culinary and concentrated medicinal use                                                                                                                                                                                                                                     | 1                                                                                                                                                                    |
| 6  | turmeric: curcumin; ginger: gingerol (or [6]-gingerol); ajwain: thymol; mint: menthol                                                                                                                                                                                                                 | 2 if all four; 1 if 2–3.                                                                                                                                             |
| 7  | New compound: shogaol. Lost: a water molecule (H <sub>2</sub> O); accept “OH group + H” or “dehydration.”                                                                                                                                                                                             | 2 if both parts; 1 if either.                                                                                                                                        |
| 8  | Heating: tulasī, haridrā, adraka, yavānī. Cooling: nimba, pudīnā.                                                                                                                                                                                                                                     | 2 if all six correctly placed; 1 if 4–5; 0 otherwise. (Note: neem’s cooling is functional, not sensory — but both major Ayurvedic classifications place it as śīta.) |
| 9  | Concentrated neem is hepatotoxic — case reports document acute liver injury, particularly with neem oil and especially in children. Organ: <b>liver</b> .                                                                                                                                             | 2 if liver named correctly + hepatotoxicity mentioned; 1 if either alone.                                                                                            |
| 10 | (a) yavānī/ajwain; thyme-like smell + ridged seed. (b) haridrā/turmeric; bright orange interior + yellow stain. (c) tulasī; toothed leaf + clove-like smell + square stem. (d) nimba/neem; compound leaf + bitterness + sickle leaflets. (e) pudīnā/mint; cooling smell + square stem + serrate edge. | 5: 1 mark per correct herb-and-feature pair.                                                                                                                         |
| 11 | A strong answer mentions: (a) turmeric is a safe culinary spice ✓; (b) “cures cancer” or “boost immunity” are overclaims relative to the evidence ✗; (c) curcumin alone is poorly absorbed — tradition pairs it with pepper to fix bioavailability; (d) honest restatement: “turmeric                 | 7: 1 for the “reasonable” part; 2 for identifying overclaim; 2 for bioavailability + tradition’s fix; 1 for honest restatement; 1 for clarity.                       |

| Q | Answer                                                                                                                                                 | Mark scheme |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
|   | is a safe culinary spice;<br>modest mechanism evidence<br>for anti-inflammatory action;<br>specific health claims remain<br>mixed in clinical trials.” |             |

**Total:** 25 marks.

**Grading bands** (for project combined out of 40): - 32–40: Exceeds - 24–31: Meets - 16–23: Approaches - 0–15: Beginning

## Homework

### Homework — Herbs — Auṣadhi (Middle)

Light daily tasks. Each takes 10–15 minutes.

1. **Day 1 — What is Auṣadhi?** · Write 2 sentences about today’s concept. (~10 min)
2. **Day 2 — Tulsī — Holy Basil** · Write 2 sentences about today’s concept. (~10 min)
3. **Day 3 — Neem — Bitter Healer** · Write 2 sentences about today’s concept. (~10 min)
4. **Day 4 — Turmeric — Golden Root** · Write 2 sentences about today’s concept. (~10 min)
5. **Day 5 — Ginger — Warming Root** · Write 2 sentences about today’s concept. (~10 min)
6. **Day 6 — Ajwain — Carom Seed** · Write 2 sentences about today’s concept. (~10 min)
7. **Day 7 — Mint — Cooling Leaf** · Write 2 sentences about today’s concept. (~10 min)
8. **Day 8 — Herbs and the Doṣas** · Write 2 sentences about today’s concept. (~10 min)
9. **Day 9 — Build the Class Deck** · Write 2 sentences about today’s concept. (~10 min)
10. **Day 10 — Final Share and Assessment** · Write 2 sentences about today’s concept. (~10 min)

## Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.



# Writing Prompts

## Writing Prompts — Herbs — Auṣadhi (Middle)

Use these for in-class writing, homework, or assessment. Each prompt is suitable for a 200-word response.

1. In 3–4 sentences, explain Herbs — Auṣadhi to a friend who hasn't taken this module.
2. Choose one Sanskrit term from this module. Define it, then describe one observation from your own life that matches.
3. Compare Herbs — Auṣadhi with a similar concept from another module or from your science textbook. What's similar? Different?
4. Identify ONE claim in this module that you'd want to test. How would you test it?
5. What was the most surprising thing in this module? Why did it surprise you?
6. Pick one source cited in this module. Write 2 sentences explaining why it matters.
7. What is one common misconception about Herbs — Auṣadhi? Rebut it in 2–3 sentences.
8. Imagine you are teaching this module to a younger student. What would you say first?

### How to use

- Pick **one** prompt per day for the writing-prompts homework.
- Or assign all of them across the module, alternating between in-class and home.
- Senior students may treat any prompt as a 500-word essay topic.

## Activity 01 — Smell-Identification Challenge

**Module:** Auṣadhi · **Band:** Middle · **Day:** woven through Days 2–7 (mainly Day 7 for the assessment version) · **Time:** 15 min total

### Type

Sensory identification, individual then paired

### Goal

Each student can identify all six module herbs by smell alone, eyes closed.

### Materials per group

- 6 small opaque cups (paper cups work) per group of 4 students — one cup per herb
- Crushed/torn samples of each herb pressed into the bottom of each cup:
  - tulsi leaf
  - dried neem leaf (sealed cup — smell through small hole only)
  - turmeric powder
  - ginger powder OR a slice of fresh ginger

- ajwain seeds (whole, lightly crushed)
- mint leaf
- 1 numbered worksheet per student

## Set-up (teacher, before class)

- Label each cup with a number (1–6). Same numbering across groups, randomised so students can't compare with their neighbour.
- Keep a master key for grading.
- Cover cups with a piece of paper held by a rubber band, with a small hole pierced for smelling.

## Instructions (student-facing)

**Smell-Identification Challenge** — individual round, then pair discussion.

1. Take the 6 cups assigned to your table.
2. Smell each cup through the hole in the cover. Do **not** open the cups. Do **not** taste anything.
3. On your worksheet, write the Sanskrit name and one feature you noticed for each numbered cup.
4. After 5 minutes — turn to your pair partner and compare answers. **Don't change yours.** Just compare.
5. Then the teacher reveals the answers. Self-score on the back of the worksheet.

## Worksheet

| Cup # | Sanskrit name | One feature |
|-------|---------------|-------------|
| 1     |               |             |
| 2     |               |             |
| 3     |               |             |
| 4     |               |             |
| 5     |               |             |
| 6     |               |             |

Self-score (after reveal): \_\_\_ / 6

## Discussion (last 3 min)

- Which herb was hardest to identify? Why?
- Was there a herb you smelled and immediately knew? Why was it easy?
- Did anyone confuse two herbs? Which pair? What do they have in common?

(Common confusion: tulsi and mint, both square-stem Lamiaceae; tulsi and clove, both eugenol-dominated; ajwain and dried thyme, both thymol-dominated.)

## Looks-like / sounds-like

|                    | Looks like                                                           | Sounds like                                                                |
|--------------------|----------------------------------------------------------------------|----------------------------------------------------------------------------|
| Going well         | Students smell each cup, look thoughtful, write a few words per herb | “This smells like clove — must be tulsi” / “Sharp! Like medicine. Ajwain?” |
| Needs intervention | Students rushing through; just writing “I don’t know”                | Silence; or “I’ll just copy you”                                           |

## Safety / sensitivity

- **No opening cups.** No tasting.
- **Neem cup smell only.** No touching the inside.
- **Allergy check before activity** — students with asthma or known plant allergies can opt out of smelling specific cups.
- **Don’t put cups directly under anyone’s nose** — let students choose their own distance.

## Differentiation

- **Tier up:** A pair gets 8 cups (the 6 plus 2 confusers — sweet basil, dried thyme). They must identify which are the 6 module herbs and which are imposters.
- **Tier down:** Only 4 cups (tulsi, ginger powder, mint, ajwain) — the most distinctive smells.

## Clean-up

- Seal all cups for next class (this activity can be reused for revision).
- Teacher checks if any cup has degraded (mint dries out fast — refresh as needed).
- Worksheets stored in the project folder.

# Activity 1 — Blind Smell-Test Identification

**Module:** Herbs — Auṣadhi · **Band:** Middle · **Duration:** 30 min

## Purpose

Reinforce the day-by-day concepts of Herbs — Auṣadhi through a hands-on activity. Aligned with NEP 2020 §4.23 (experiential learning).

## Materials

small samples of each of the six herbs, blindfolds (optional), notebook

## What students do

Students close their eyes and try to identify each herb by smell. They record their guess + observation, then check.

## Step-by-step

1. **Setup (5 min)** — gather materials; explain the task.
2. **Working phase (20 min)** — students work individually or in pairs.
3. **Share (5 min)** — selected students share their results with the class.

## Differentiation

- **Extension:** ask advanced students to do an additional iteration or analysis.
- **Support:** pair students who need scaffolds; offer partial templates.

## Assessment

This activity is **formative**. The teacher observes participation, accuracy, and reflection quality. No grade is assigned.

## Safety

- General safety briefing before any materials are handled.
- If herbs, plants, or food are involved: kitchen-amount only, no ingestion of unknown preparations.
- Outdoor activities: stay in the designated area; supervised by the teacher.

## Activity 02 — Sort by *Vīrya*

**Module:** Auṣadhi · **Band:** Middle · **Day:** 8 · **Time:** 15 min (within Day 8's 45 min)

## Type

Card-sort, paired, with defended reasoning

## Goal

Each pair correctly sorts 12 kitchen items into heating (*uṣṇa*) / cooling (*śīta*) / not-sure, and articulates the **reasoning** behind their classification — using both sensory observation and the Ayurvedic logic of taste-and-effect.

## Materials per pair

- 12 cards, each with one item name (no answer printed):
  - **6 module herbs:** tulasī, nimba, haridrā, adraka, yavānī, pudīnā
  - **6 kitchen-pantry items:** rice, ghee, lemon, ice cream, red chilli, coriander seed
- 1 sorting mat with three columns labelled *Uṣṇa* (heating) / *Śīta* (cooling) / *Not sure*
- 1 worksheet per pair (below)

## Set-up (teacher, before class)

- Pre-shuffle each pair's 12 cards.
- Have the sorting mats ready on each desk.

## Instructions (student-facing)

**Sort by *Vīrya*** — work in pairs.

1. Take your 12 cards and your sorting mat.
2. For each item, decide together: heating (*uṣṇa*), cooling (*śīta*), or not sure.
3. Fill in the worksheet — name each item, your choice, and **one sentence explaining why**.
4. **Disagree with your partner?** Argue it out before you fill in the row. Use sensory evidence (“ginger feels warm when I eat it”) or the doṣa logic (“bitterness reduces pitta, so neem is functionally cooling”).
5. Last 3 min: gallery walk — visit two other pairs’ mats. Note one card they placed differently from you.

## Worksheet (one per pair)

| #  | Item           | Heating / Cooling<br>/ Not sure | Reasoning (one<br>sentence) |
|----|----------------|---------------------------------|-----------------------------|
| 1  | tulasī         |                                 |                             |
| 2  | nimba          |                                 |                             |
| 3  | haridrā        |                                 |                             |
| 4  | adraka         |                                 |                             |
| 5  | yavānī         |                                 |                             |
| 6  | puḍīnā         |                                 |                             |
| 7  | rice           |                                 |                             |
| 8  | ghee           |                                 |                             |
| 9  | lemon          |                                 |                             |
| 10 | ice cream      |                                 |                             |
| 11 | red chilli     |                                 |                             |
| 12 | coriander seed |                                 |                             |

## Teacher key (do not show to students until after sort)

| Item    | Classification             | Note                                         |
|---------|----------------------------|----------------------------------------------|
| tulasī  | <i>uṣṇa</i> (heating)      | Eugenol-<br>dominated<br>aromatic<br>warming |
| nimba   | <i>śīta</i> (cooling)      | Functional:<br>bitterness<br>reduces pitta   |
| haridrā | mild <i>uṣṇa</i> (heating) | Curcumin; some<br>texts call it<br>neutral   |
| adraka  | <i>uṣṇa</i> (heating)      | Gingerol                                     |

| Item           | Classification        | Note                                                                                                |
|----------------|-----------------------|-----------------------------------------------------------------------------------------------------|
|                |                       | activates TRPV1 (warm)                                                                              |
| yavānī         | <i>uṣṇa</i> (heating) | Thymol pungent stimulant                                                                            |
| pudīnā         | <i>śīta</i> (cooling) | Menthol activates TRPM8 (cool)                                                                      |
| rice           | <i>śīta</i> (cooling) | White rice in classical texts; basmati specifically <i>śīta</i>                                     |
| ghee           | <i>śīta</i> (cooling) | Classical Ayurvedic classification — interesting because it’s warming sensation but cooling potency |
| lemon          | <i>uṣṇa</i> (heating) | Sour and heating in classical texts (some debate; some texts call it neutral)                       |
| ice cream      | <i>śīta</i> (cooling) | Both sensory and functional                                                                         |
| red chilli     | <i>uṣṇa</i> (heating) | Capsaicin activates TRPV1                                                                           |
| coriander seed | <i>śīta</i> (cooling) | One of the few <i>śīta</i> spices — used to balance heat                                            |

**Note for the teacher:** the “interesting” ones are **ghee** (warming feeling but cooling potency in classical texts) and **lemon** (sour and somewhat heating despite cooling lemonade culture). These are where students will argue most productively. The argument is the learning.

### Discussion prompts (last 3 min, whole class)

- Which item caused the most disagreement in your pair?
- Did anyone classify ghee as cooling? Why might that be?
- What’s the difference between “feels warm/cool” and “has a warming/cooling effect over time”? (This is the sensory vs functional point.)

## Looks-like / sounds-like

|                    | Looks like                                                                                  | Sounds like                                                                                                   |
|--------------------|---------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Going well         | Two partners pointing at cards, both holding the sorting mat, sometimes laughing about ghee | “But ghee feels warm!” / “Yes but it cools pitta...” / “Maybe both? Let’s go with cooling but write ‘tricky’” |
| Needs intervention | One partner sorting silently; cards being placed without discussion                         | Silence or “I don’t care, put it anywhere”                                                                    |

## Safety / sensitivity

- No food handling in this activity (it’s card-only).
- If a student raises dietary or religious restrictions about any item (e.g. ghee, ice cream), acknowledge and continue — the sort is a classification exercise, not a food choice.

## Differentiation

- **Tier up:** Add 4 more cards — *aśvagandhā* (heating; reduces vāta), *brāhmī* (cooling; pitta-reducing), green tea (cooling), black pepper (heating). Pair must research one of the unfamiliar items before classifying.
- **Tier down:** Use only 8 cards (the 6 herbs + ice cream + chilli) — the most clearly classifiable.

## Clean-up

- Cards collected and stored in card bag for future use.
- Worksheets stapled into the student workbook.

# Activity 03 — Herb Deck Build

**Module:** Auśadhi · **Band:** Middle · **Day:** 9 (and finalised Day 10 morning) · **Time:** 35 min in-class + variable at home

## Type

Project artefact construction, paired

## Goal

Each pair produces six A6 (or quarter-A4) cards — one per module herb — with all eight required elements per card. The deck becomes the pair’s project submission and a classroom resource after Day 10.

## Materials per pair

- 8 blank cards (6 + 2 spares)

- Colour pencils, fine-tip markers, ruler
- Reference materials (their own Days 2–8 worksheets)
- 1 envelope or rubber band for storing the finished deck

## The card template (eight elements per card)

|                       |       |
|-----------------------|-------|
| [ SKETCH OF HERB ]    |       |
| Sanskrit (IAST):      | _____ |
| English / Common:     | _____ |
| Botanical name:       | _____ |
| Family:               | _____ |
| Identifying features: |       |
| 1.                    | _____ |
| 2.                    | _____ |
| 3.                    | _____ |
| Active compound:      | _____ |
| Traditional use:      | _____ |
| Safety note:          | _____ |

## Instructions (student-facing)

**Today: build the deck.**

1. Lay out your six blank cards. Decide who draws which (3 each, then swap and review).
2. Fill in the eight elements per card. Use your worksheets — don't re-invent.
3. **Sketch first** (least time-consuming), **safety note last** (most time-consuming to get right).
4. Check each card against the rubric before finalising. The rubric is on the wall.
5. By the end of class, all six cards should be drafted. Refine at home tonight.

## Quality checklist for each card (rubric-aligned)

- ☐ Sketch is clear enough that someone who's never seen the herb could recognise it
- ☐ Sanskrit name in IAST (with diacritics)
- ☐ Botanical name italicised, correctly spelled
- ☐ Family named (one of: Lamiaceae, Apiaceae, Zingiberaceae, Meliaceae)
- ☐ Three distinct identifying features (not three versions of the same feature)
- ☐ Active compound correctly named (eugenol, azadirachtin, curcumin, gingerol, thymol, menthol)
- ☐ One specific traditional use (no vague claims)
- ☐ One specific safety note (no vague reassurances)

## Common failure modes (teacher watches for these)

1. **Sketching one card for 20 minutes.** Cap at 5 minutes per sketch.



2. **Vague traditional use** (“good for health”). Demand specificity: “*good for what — relieving congestion? helping digestion? cooling in summer?*”
3. **Vague safety note** (“be careful”). Demand specificity: “*be careful of what — concentration? dose? interactions?*”
4. **Two partners not coordinating**. If you see this — have them switch sides at minute 15.

## Differentiation

- **Tier up:** A 7th bonus card (any herb), applying the same eight-element template — *aśvagandhā, brāhmī, triphalā, kariveppilai, dhānyaka*. Plus: write a 30-second pitch for the bonus card on the back.
- **Tier down:** Reduce to 4 cards (tulsi, ginger, turmeric, mint) but each must have all 8 elements complete. Quality over quantity.

## Looks-like / sounds-like

|                    | Looks like                                                                                         | Sounds like                                                                             |
|--------------------|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Going well         | Both partners drawing/writing; checking the rubric on the wall; quick discussion before each entry | “What’s the safety note for ginger?” / “Heartburn at high doses?” / “Yes — write that.” |
| Needs intervention | One partner doing everything; cards being copied verbatim from worksheets without engagement       | Silence; or “let’s just write ‘good for cold’”                                          |

## Clean-up

- Each pair stores their cards in an envelope with both names + class on the front.
- Refine at home overnight; bring on Day 10.

## Project Brief

### Project Brief — Class Herb Identification Deck

**Module:** Herbs — Auśadhi · **Band:** Middle

## What you will do

Each student creates one identification card for one of the six herbs: Sanskrit/botanical name, leaf drawing, smell description, three traditional uses, one active compound, one safety note. Cards combined into class library.

## Why

This project asks you to apply concepts from this module to your own life or a real situation. It is graded on the rubric in `rubric.md`.

## Deliverables

- **One-page project plan** (Day 2 of the module): problem, sources, deliverable, timeline.
- **Daily journal entries** (one per day, starting Day 6): what you did, what worked, what didn't.
- **Final artefact** (a poster / model / digital deliverable).
- **Written reflection** (300 words (Middle)): what you learned, what surprised you, what you're still uncertain about.
- **Presentation** (5-minute share with the class).

## Timeline

- **Day 1–2:** scope and plan.
- **Day 3–4:** research.
- **Day 5:** mid-project critique.
- **Day 6–7:** build / make.
- **Day 8:** document.
- **Day 9:** rehearse.
- **Day 10:** share day.

## Sources

You must cite at least 2 sources. Use the form: *Author, Title, year*. (See `../sources.md` for examples.)

## Safety

- Any project involving plants / food / outdoor work must be supervised.
- **No herbal medicine recommendations.** Modules 10 and 4 give awareness; therapeutic decisions belong to qualified practitioners.

## Grading

See `rubric.md`. The project is worth the same weight as the summative quiz.

## Rubric

### Project Rubric — Module 10 Auṣadhi (Middle)

**Project marks:** 15 (combined with the summative quiz's 25 for a module total of 40).

## Criteria

| Criterion                                                             | 4 — Exceeds                                                                                                                                                                                                      | 3 — Meets                                             | 2 — Approaches                                                                          | 1 — Beginning                                       | Weight |
|-----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------------------------------------------------|-----------------------------------------------------|--------|
| <b>Conceptual accuracy (active compound, family, traditional use)</b> | All six cards have correct Sanskrit + botanical + family + active compound. Plus: traditional uses are <i>specific</i> (e.g. “ajwain water for indigestion”), not vague.                                         | 5–6 cards fully accurate; minor errors on 1 card.     | 3–4 cards fully accurate; rest have meaningful errors (wrong compound, wrong family).   | Fewer than 3 cards accurate.                        | × 2    |
| <b>Safety framing</b>                                                 | Each card has a specific safety note — not vague (“be careful”) but specific (“neem oil is hepatotoxic; do not ingest”). The neem card is clear and the essential-oil warning is explicit somewhere in the deck. | Each card has a safety note; most are specific.       | Some cards have safety notes; some vague; the neem warning is present but not specific. | Safety notes vague or missing.                      | × 2    |
| <b>Identification features (visual + sensory)</b>                     | Each card has 3 <i>distinct</i> features — leaf shape, smell, family characteristic, etc. Sketches are accurate enough that a stranger could identify the                                                        | Each card has 3 features; most sketches are accurate. | Features sometimes redundant (“green leaf, has a smell”); sketches simplified.          | Features missing or vague; sketches unrecognisable. | × 1    |

| Criterion                           | 4 — Exceeds                                                                                                                                                                                          | 3 — Meets                                                                 | 2 — Approaches                                                          | 1 — Beginning                                       | Weight |
|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|-------------------------------------------------------------------------|-----------------------------------------------------|--------|
|                                     | herb.                                                                                                                                                                                                |                                                                           |                                                                         |                                                     |        |
| <b>Connection to doṣa framework</b> | At least one card explicitly notes the herb's <i>vīrya</i> (heating/cooling) AND links it to a doṣa ( <i>vāta</i> / <i>pitta</i> / <i>kapha</i> ) effect.                                            | Some cards note heating/cooling; doṣa link present somewhere in the deck. | Heating/cooling mentioned vaguely.                                      | No mention of <i>vīrya</i> or doṣa.                 | × 1    |
| <b>Source citation</b>              | The deck references the <i>Auśadhi-sūkta</i> (RV 10.97) or the <i>Caraka Saṁhitā</i> (Sūtrasthāna) at least once, accurately. Plus: at least one phytochemistry reference (named compound) per card. | Citation present but only at one level (scripture or compound, not both). | Citation vague (“ancient texts say...”).                                | No citation.                                        | × 1    |
| <b>Communication / craft</b>        | Cards are visually clean, IAST diacritics correct, italicisation correct for botanical names, handwriting legible.                                                                                   | Clear communication; one or two craft issues.                             | Some craft issues — multiple typos, missing italics, illegible patches. | Hard to read; multiple issues impede understanding. | × 1    |

## Score calculation

Project score = (sum of criterion scores × weights) × (15 / 32)

Maximum raw score: 32. Multiply by 15/32 = 0.46875 to get out of 15.

## Self-reflection (folded into project score)

The reflection slip (submitted with the deck) asks three questions. If the reflection is missing or perfunctory, the **Safety framing** mark drops by 1 (this is the most important takeaway of the module, and a perfunctory reflection suggests it didn't land).

## Common decision points

- **Group of 3 vs pair:** Acceptable. The third member must contribute at least 2 cards. Marks are equal across the group unless one member visibly didn't contribute.
- **Bonus 7th card:** Bonus card adds up to +1 mark to the **Conceptual accuracy** criterion (capped at 4). Doesn't change the rubric scale.
- **Late submission:** -1 mark per school day, capped at -3.
- **Plagiarism / web copying:** If a card is closely copied from a web source not cited, the deck goes to 0 on **Source citation** AND on **Conceptual accuracy**. A 1-day extension is offered for a redo.

## Marking process

- Mark in real-time during the Day 10 presentation (2 min present + 30 sec Q&A + 30 sec marking).
- Write one sentence of feedback per pair after their presentation.
- Return marked decks (or photos of them, if archived) within 2 school days.

## Note on the “Exceeds” tier

A deck reaches Exceeds when it's not just complete but **useful**. The clearest signal: another teacher could pick up the deck and use it to teach a 5-minute lesson on any of the six herbs. If you find yourself wanting to borrow the deck for the staff room, that's 4.